An encyclopedia of the factor bases we have tried

Date: 2026-10-15. **Scope:** every factor-base construction for elliptic-curve and Jacobian index calculus that this repository (`aburan28/crypto`, at `a80b15bbc`) or the research program (`aburan28/crypto-autoresearcher`, at `4446b91c2f`) has designed, built, measured or closed on paper. The two sibling repositories `aburan28/cryptanalysis` and `aburan28/ml-cryptanalysis` are included where they hold a family that is not mirrored here. **Kind of document:** a search round over code, notes and records (AGENTS.md "Research searches must leave visual reports"). It compiles results that are already recorded and runs nothing new. Every number below is quoted from the file cited next to it. Numbers marked derived here are this note's own arithmetic on quoted numbers, and the working is shown. **Corrected:** 2026-10-16 (claim audit): §0.1 and §8 lead 1 now cite `research/ic_next10_20261014/03-pkm-degree` (`D ≥ 8` at `m = 4, N = 20`, NEGATIVE), and the P-192 `t = 48` card is marked superseded. **Updated:** 2026-10-15, with the outcomes of the five lead studies that merged to `main` at `b906df024` (PRs #1887, #1889, #1892, #1893, #1895). They were chased from §8 of the first version; each verdict below is quoted from the study's README on `main`, and the earlier text is corrected where a study superseded it. **Updated again:** 2026-10-16, at `main 05e414f3f`. This round adds the studies merged since PR #1916: the P-192 auxiliary-curve search (PR #1925), the supersingular auxiliary curve (PR #1947), the generic-field F4 engine at NIST primes (PR #1942), the frontier campaign's band-priced ratios (PR #1837) and the binary MITM oracle ladder (PRs #1919, #1944, #1964). It also adds entries that the first two versions missed: the N83 Koblitz programme (D8), the large-prime domain (D9), solver-cost selection (D10), the controlled coset comparison (A13) and the genus-3 cover after cubic GHS (E6). Every number added in this round was checked against the cited study on `main`. **Report PDF:** `REPORT.pdf`, built from this file. **Figures:** `figures/taxonomy.svg` and `figures/cost_vs_rho.svg` map the families; `figures/pkm_solving_degree.svg` plots the two flagship families. The data is in `figures/data.json` and the drawing code in `figures/plot.py`.

Path conventions:

- a bare path is in `aburan28/crypto`;
- `AR:` marks `aburan28/crypto-autoresearcher`;
- `CA:` marks `aburan28/cryptanalysis`;
- `ML:` marks `aburan28/ml-cryptanalysis`.

`S` is the repository's cost unit: group-addition equivalents divided by `√r`. Rho's `S` is about 1.25–1.3 without the negation map and 0.886 with it. "× rho" always means the ratio the cited source recorded against its own matched rho arm.

These ratios come from different curve sizes and different accounting, so two rows are not comparable unless the text says they are.

Pricing weight (AGENTS.md §5a). Since 2026-10-15 a cross-method ratio or ranking must price every arm's unpriced native work (lookups, row operations, canonicalisations and the like) at 0, 0.1 and 1 group-addition equivalents per operation, and a ranking stands only if it holds at every weight. Every "× rho" in this note that does not say otherwise was recorded before that rule, at weight 0 with the unpriced work left out on both sides. Read it as a **lower-bound comparison**, not a verdict. Ratios that were priced across the band say "band" and give the weights. Wall-clock ratios (the N83 rows, D8) say "wall"; they are not in the `s` unit and were not band-priced.

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0. The two flagship families, and what was actually run on them

The prime-field factor bases that matter most for deployed curves are Petit, Kosters and Messeng's tower bases (PKC 2016). Their factor base is

```
F = { P ∈ E(F_p) : x(P) ∈ V },   V = L⁻¹(c),   L = L_t ∘ … ∘ L_1,
```

where each `L_j` has low degree and the preimage tree of `L` splits completely over `F_p`. Then `|V| = 2^t` (for degree-2 steps). Membership of `x` in `V` is written as `t` low-degree equations in auxiliary variables `y_0 = x, y_1, …, y_{t−1}`:

```
T_j : y_{j+1} − L_{j+1}(y_j) = 0        (0 ≤ j < t−1)
T_t : L_t(y_{t−1}) − c = 0
```

so no membership equation has degree above 2, even though `V` has `2^t` elements. This is the prime-field stand-in for the `F_2`-subspace of binary Weil descent. The repository's design is in `RESEARCH_PKM_TOWER_ORACLE.md` (the PKM note), §2.1, lines 205–254, fixed in commit `c2157282` before any run.

Two of its tower kinds correspond to the two factor-base shapes PKM's slides attach to NIST curves:

	0.1 Kummer: a coset of a smooth subgroup of `F_p^×`	**0.2 Isogeny: x-coordinates from an isogeny chain on an auxiliary curve**
NIST curve it fits	**P-224**: `p − 1 = 2^96·(2^128 − 1)`, so `v₂(p − 1) = 96`	**P-192**: `v₂(p − 1) = 1`, so no Kummer tower exists; `v₂(p + 1) = 64`

V	$g \cdot \mu_{2^t}$, the coset of the 2^t-th roots of unity through g	$x(R' + \langle Q \rangle)$ for a point Q of order 2^t on an auxiliary curve E'/F_p
step map	$y \mapsto y^2$, with $y_0 = x/g$ and $y_{t-1}^2 = 1$	the x-map of a 2-isogeny (Vélu): $y_{j+1}(y_j - \xi_j) = y_j^2 - \xi_j y_j + \tau_j$
existence	$2^t \mid p - 1$	none on p: E' is searched for
build cost	one exponentiation for ζ, then 2^t multiplications	an auxiliary-curve search, about 2^t point counts, charged to the factor-base phase
implementation	`prime-kummer:t=T[,g=G]` in `src/cryptanalysis/ic_framework/prime_tower.rs` (an `ic.pipeline` plug-in), and `examples/pkm_tower_pilot.rs`	`prime-isogeny:t=T[,seed=S]` in the same file since PR #1889 (u64, $t \le 20$, search charged), with `aux=supersingular` since PR #1947; `examples/pkm_tower_pilot.rs` (lines 497–590) for the solver runs; `pkm-aux-search` and `src/cryptanalysis/two_sylow_search.rs` for the P-192 auxiliary curve (PR #1925)
largest instance run	solver: $t = 11$ ($N = m \cdot t \le 22$) at $p_1 = 2013265921$; Kummer at $v_2(p - 1) = 40, 48, 56$ and at the P-224 prime up to $m = 2$, $N = 20$ and $m = 4$, $N = 12$ (PR #1942)	solver: $t = 10$ ($N \le 20$) at $p_0 = 786433$, and with a supersingular E' up to $N = 20$ at 24–31-bit primes (PR #1947); ecbench: $t = 8\ldots11$ at 20–28 bits (PRs #1889, #1947); construction: E' with a point of order 2^{38} (ordinary) and 2^{48}, 2^{64} (supersingular) at the P-192 prime, and V

		built there for `T ≤ 16` (PR #1925)
run at NIST size?	**F4 solver only**: `D`, width and steps at the P-224 prime equal the 31–32-bit values (PR #1942). No `S`, no relation phase	**Construction only**: the auxiliary curve and the tower `V` at the P-192 prime (PR #1925). No F4 run and no relation phase at P-192

What has run on a NIST prime (since 2026-10-16). Until PR #1942 every tower run used a generated prime of 20–32 bits, because the F4 tower engines stored coefficients in 32-bit words and asserted `p < 2³²` (PR #1893). The generic-field engine `f4_tower_generic` (`research/f4_generic_field_20261015/README.md`) now runs the Kummer tower at the P-224 prime and the Dickson torus tower at the P-192 and P-256 primes. It gives the recorded small-prime `D`, width and step count in all 82 cells run (§0.1). The isogeny tower has an auxiliary curve and a built `V` at the P-192 prime (PR #1925, §0.2), but **no F4 run at P-192 yet**: its system builder is still `u64`. Both are stage diagnostics of the solver or of the construction. Neither family has a priced relation phase (`S`, ratio to rho) at a NIST size; the `prime-kummer` and `prime-isogeny` plug-ins work in `u64` (`prime_tower.rs`, and `primitive_two_power_root` refuses `t ≥ 63`). The NIST primes' 2-adic valuations are in the PKM note §10.10 (lines 1142–1162), "properties of the primes, not measurements", and PKM's slide remark is quoted in `research/pdp_root_finder_requirements_20261012/README.md` lines 104–119: "NIST P-224 has `p − 1 = 2^{96}·N'`". The autoresearcher ledger has no P-224 coset base and no P-192 isogeny-chain base. Its only P-192 factor-base object is a class-group base, entry E3 below. `research/elliptic_snfs_survey_20261016/README.md` (PR #1951) places both towers as shape C of an "elliptic SNFS": the only prime-field analogue of a special factor base, with open asymptotic cost, and one whose isogeny variant works for every prime.

0.1 Kummer tower: `g·μ_{2^t}` (the P-224 shape)

Definition. Fix `t` with `2^t | p − 1`, a primitive `2^t`-th root of unity `ζ`, and `g ∈ F_p^×`. Then `V = g·μ_{2^t} = {g·ζ^k : 0 ≤ k < 2^t}`, and `F = {P ∈ E(F_p) : x(P) ∈ V}`. About half of `V` lifts to points, and both signs are kept. The tower is `y_0 = x/g`, `y_{j+1} = y_j²`, and `y_{t−1}² = 1`.

Algebra (PKM note §2.4, Lemma 1, lines 296–319). The tower equations are a Gröbner basis with square-free standard monomials, and the quotient ring is `R = F_p[y]/(towers) ≅ F_p^{V^m}`. For Kummer, `R` is the group algebra `F_p[(Z/2^t)^m]`, and `x_i^e = g^e·∏_{j ∈ bits(e)} y_{i,j}`.

What theory predicted (§3.2–3.4, lines 336–418).

- A PKM system has the field-equation half of Weil descent, a finite quotient with a square-free monomial basis. It lacks the conjugate-equation half, which needs a Galois group that F_p does not have. So it is a one-generator system over a 2^N-point quotient, $N = m \cdot t \approx \log_2 r$.
- Proposition 2: the XL (Macaulay) solving degree is at least $d + \delta$, where δ is generically about N.
- Kummer and Dickson towers are degenerate at infinity, with Bardet–Faugère–Salvy degree of regularity $N + 1$. Isogeny towers are not.
- §3.7, the linear-algebra floor: the optimum costs about $r^{2/(m+1)}$, so $m \le 3$ can never beat rho. $m = 4$ gives $r^{2/5}$ and $m = 5$ gives $r^{1/3}$, so "the decisive Stage A cells are $m \ge 4$".

The pre-registered prediction failed in the bounded direction at first. F4's solving degree D did not grow like N. Later rounds showed it is not bounded either. The `f4_fp_tower` results, all Kummer at $p_1 = 15 \cdot 2^{27} + 1$, are consolidated from `research/pdp_root_finder_requirements_20261012/README.md` lines 238–255:

m	N	D	width (columns)	F4 multiply-adds	peak mem-ory	F4 time
2	12	5	1,231	6.50e7	8 MB	0.05 s
2	16	5	4,789	6.18e9	36 MB	1.1 s
2	18	5	8,644	4.60e10	91 MB	6 s
2	20	**6**	33,446	4.84e11	473 MB	57 s
2	22	6	51,909	—	2.3 GB	10 min
3	9	6	1,186	1.59e8	11 MB	0.1 s
3	12	6	5,019	1.06e10	64 MB	2.0 s
3	15	**7**	31,397	3.19e12	1.1 GB	6.0 min
3	18	7	106,288	8.99e13	12.7 GB	176 min
4	8	6	1,049	4.84e7	9 MB	0.1 s
4	12	**7**	9,452	8.55e10	171 MB	9.4 s
4	16	7	76,837	2.7e13	8.1 GB	55.5 min
4	20	**≥ 8** (msolve)	—	—	5 GB cap	—

The m = 4, N = 20 row is msolve 0.6.5 on the raw presentation, not f4_fp_tower, from research/ic_next10_20261014/03-pkm-degree/ (2026-10-14; see the correction under §0.1 Status).

The rounds and controls behind that table:

- **Pilot, §10.2** (2026-09-24, p_0 = 786433 = 3·2^18 + 1). m = 2, N = 2…18 gives D = 4,4,4,4,4,5,5,5,5.
  - The naive presentation, one dense degree-2^t membership polynomial, gives exactly D = 2^t + 2: 66 against 5 at N = 12. The tower presentation is what keeps the degree low.
  - A shape-matched **null** tower gives the same D, width and steps, so the low degree comes from the tower algebra and its monomial support, not from the curve.
  - dickson gives the same D, width and step count as kummer at every N. It is the same F4 computation on different numbers, so it is not an independent replication.
- **Round 2, §11** (2026-09-25): D rises to 6 at N = 20, 22 for m = 2, in the Kummer and the isogeny family and in the null. This refuted the pilot's Conjecture 3 (D ≤ 5 for all t).
- **Round 3, §12**: m = 4, N = 16 gives D = 7 on both targets in 55 minutes and 8.1 GB. From N = 12 to N = 16, the width grew ×8.1 and the multiply-adds ×316.
- **Round 4, §13**: a signature engine (F5/GVW, sig_fp_tower) needs rows 1–2 degrees higher than F4 on 18 of 18 systems, so F4 stays the instrument.
- **Round 6, §15**: an early exit cuts 39% of the multiply-adds at m = 4, N = 12. This is engineering.
- **Round 7, §16** (2026-10-06): m = 3, N = 18 gives D = 7, so the m = 3 line reads 6, 6, 7, 7. N = 21 needs more than this host's 14 GB.
- **IC next-10, item 3** (2026-10-14, research/ic_next10_20261014/03-pkm-degree/): **NEGATIVE for the m = 4 plateau.** At m = 4, N = 20, on both targets, msolve completes three productive degree-8 F4 rounds before its 5 GB cap; it reduces every lower-degree pair first, so **D ≥ 8**, and the m = 4 line reads D = 6, 7, 7, ≥ 8. m = 3 stays inconclusive (D ≥ 7 at N = 18, 21, 24, stopped by the wall before any degree-8 round). Caveat: msolve on the raw presentation; f4_fp_tower has not run N = 20.

Pre-registered reading (rule A1, lines 1506–1513 and 3169–3171):

cell	D sequence	reading
m = 2	5 5 5 5 6 6	"H0, at the threshold"

m = 3	6 6 7 7	inconclusive
m = 4	6 7 7	inconclusive

log₂(multiply-adds) grows 1.6, 2.1 and 2.4 bits per tower variable at m = 2, 3, 4, where plain enumeration grows 0.50, 0.67 and 0.75: "F4 grows three to seven times faster than plain enumeration" (requirements README lines 260–266).

What a deployed size would require. The requirements README §6.4 (lines 420–438) works this out for a 256-bit p at m = 4: t = 64, N = 256, T_dec ≲ p^{1/4}/32 ≈ 2^59. "Admissible degree: 3 at m = 4, 4–5 at m = 5; measured: 7 at m = 4, N ≤ 16, rising." Derived here: for P-224 at m = 4, the same sizing gives t ≈ 56 (N ≈ 224). That is well inside v₂(p − 1) = 96, but 52 levels past the largest t = 4 measured at m = 4.

End to end in ecbench (research/pkm_tower_ecbench_20261013/README.md). The prime-kummer:t=8…11 base was run on five curves of 20–28 bits with 2-adicity 12, with every group operation charged. 896 runs, all verified.

bits	rootfind ÷ rho (S, 95% CI)	tower ÷ plain base of equal size
20	34.0 [27.9, 42.6]	1.006 [0.982, 1.032]
22	41.9 [34.3, 53.4]	1.001 [0.996, 1.006]
24	29.4 [23.7, 37.3]	1.012 [0.989, 1.035]
26	96.6 [77.8, 124.1]	1.005 [0.994, 1.020]
28	140.6 [113.0, 182.3]	0.999 [0.994, 1.004]

"PKM's structure helps an algebraic solver, not a table." The ratio grows 0.26 ± 0.28 bits per bit of r. Class: engineering.

Related falsifier and root-finding results.

- **C4 falsifier** (research/ic_exponent_falsifiers_20261013/README.md): is e_a + e_b + e_c mod 2^k biased on the t = 8 base at p = 946177? It is not. The largest class share is 0.680, 0.508, 0.393 … against 0.680, 0.511, 0.396 … for the null, while the positive control reads 1.000. "C4's simplest form is falsified on this base."
- **Root-finding oracles on the same cells** (research/pdp_root_finder_20261012/README.md): they were exact on 707 of 707 brute-force checks, with $4.6 \cdot 10^{11}\times$ less counted work than F4 at m = 3, N = 18. Their slopes sit on the pair-table and enumeration floors. Class: engineering.

Autoresearcher siblings.

- Engine sweep (AR:src/crypto_autoresearcher/index_calculus/README.md). It uses a high-degree, non-tower subgroup base (x ∈ g·μ_d, entry A4b) and measures exponent 0.97 [0.95, 0.99] at m = 2, the same as small_x.
- Twisted group algebra (AR:IDEA-20260808-9ef88c, proposed). Its Stage-0 analysis predicts solving degree B − 1 for the binomial membership x^B − c^B, and a cost Õ(B^m), "still worse than MITM at theta = ceil(m/2)".

Deeper 2-adicity, the torus variant, and the cost of the next cells (2026-10-15, PR #1893, research/pkm_dickson_torus_20261015/README.md).

- Deeper 2-adicity (measured). At $p_2 = 3\cdot 2^{30} + 1$ ($v_2(p-1) = 30$), Kummer D and width equal the recorded values at p_0 ($v_2 = 18$) and p_1 ($v_2 = 27$) in every (m, N) cell run (m = 2, N = 2…18; m = 3, N = 6…12), and so do the steps wherever the recorded targets were also random. The study derives why: ζ enters only through V, which the system never contains. $v_2 \geq 40$ could not be run there: both F4 engines held 32-bit coefficients, and no prime below 2^{32} has $v_2(p-1) > 30$. PR #1942 (below) removed that limit.
- The Dickson torus tower (2^t | p + 1) gives the split D in all 16 cells; see 0.2 and A5c.
- The next cells, derived and not run. m = 3, N = 21 projects to about 130–150 GB peak memory and 2.5–7.4·10^{15} multiply-adds; m = 4, N = 20 to about 240–380 GB and 0.85–2·10^{16}. Each needs a 192–256 GB or a 512 GB host, or a disk-backed reducer block. A value at m = 3, N = 21 could extend the D = 7 plateau to L = 6 or show a rise; neither gives H1a, which needs N ≥ 27 at m = 3.

The tower at NIST primes: a generic-field F4 engine (2026-10-15, PR #1942, research/f4_generic_field_20261015/README.md).

- Built. f4_tower_generic<F: PrimeField>, the u32 engine's algorithm over any field backend (Fp32, the AVX-512 IFMA Fp52, Fp64, FpP192, Fp224, FpWide<N>), with a wide system builder above 2^63. At Fp32 it reproduces f4_fp_tower field for field on 182 rows (0 differences). The 5 pre-registered speed claims pass: 1.091× and 1.058× at $p < 2^{31}$, 2.18× and 2.24× above 2^{31}, and Fp52 2.57× Fp64 at 45 bits (engineering).
- Measured, H_same in all 82 cells. Kummer at $v_2(p-1) = 40, 48, 56$ ($27\cdot 2^{40}+1$, $15\cdot 2^{48}+1$, $27\cdot 2^{56}+1$) and at the P-224 prime ($v_2 = 96$), for m = 2, N = 2…18, m = 3, N = 6, 9, 12 and m = 4, N = 8, and the Dickson torus at P-192 (m = 2, N ≤ 18; m = 3, N ≤ 12) and P-256 (m = 2, N ≤ 16; m = 3, N ≤ 9): 74 cells, each with the recorded 31–32-bit D, width and step count. Eight extension cells (one target each) agree as well. The key ones:
  - Kummer m = 2, N = 20 at P-224: D = 6, width 33,446, 24 steps, as K1; 3,690 s;
  - Kummer m = 4, N = 12 at P-224: D = 7, width 9,452, 29 steps, as M4;

  - Kummer m = 3, N = 15 at v_2 = 40: D = 7, width 31,397, as M3;
  - torus m = 2, N = 20 at P-192: D = 6, width 33,426, as T31. On the fixed binary the multiply-add count of a Kummer cell is the same at 45, 52, 61 and 224 bits (481,961,879,776 at m = 2, N = 20): "the computation does not see the prime at all at these sizes". Every finished P-192, P-224 and P-256 row passed the wide builder's BigUint re-check (--verify, 82/82), and every u64-builder row an exhaustive recount (97/97).
- Cost per multiply-add (descriptive, shared host): 14.47 ns at P-224 (Fp224) and 8.51 ns at P-192, against 0.479 ns for Fp32 at p_1.
- What it settles. Lead 4 (§8): D does not depend on $v_2(p - 1)$ or on the size of the prime, up to P-224, wherever it was measured. What it does not settle. Whether D stays bounded as N grows (lead 1); every cell is at N ≤ 20, and the P-224 statement is a solver-stage diagnostic with no S.

Priced across the §5a band (2026-10-15, PR #1837, research/ic_frontier_campaign_20261015/README.md). The frontier campaign re-ran the prime-kummer tower in ecbench at 20–28 bits ($\log_2$ r 18.9–26.2) and priced every arm's unpriced work at 0, 0.1 and 1 GAE per native operation. The tower arm's mean S over the negation-map rho rises from **30.3 to 141.3×** across the sizes at w = 0, 30.4 → 158.7× at w = 0.1 and **31.0 → 238.7×** at w = 1. Pricing the unpriced work widens the gap, so the ranking (rho first) holds at every weight. The campaign's verdict covers every family it ran: "No index-calculus arm beats generic rho on any curve at any band weight; no index-calculus arm has a fitted cost exponent below rho's."

Status. Correction (2026-10-16 claim audit): the earlier status said "at m ≥ 4 there is no data past N = 16". That missed research/ic_next10_20261014/03-pkm-degree/, which measured **D ≥ 8 at m = 4, N = 20 (NEGATIVE)**. So the solver axis is negative at m = 4, the only regime §3.7 allows to beat rho's exponent, and inconclusive at m = 3, whose next cell needs a large-memory host (above). The P-224 cells reach N = 12 at m = 4. 2-adicity up to v_2 = 96 (the P-224 prime) changes nothing in D. End to end it is engineering: a tower base costs the same as a plain base in a table oracle, and 30–239× rho across the band at 20–28 bits. No relation phase has run at P-224 size.

0.2 Isogeny tower: x(R' + ⟨Q⟩) on an auxiliary curve (the P-192 shape) Definition (PKM note lines 222, 233–242).

- Search auxiliary curves E'_0 = E'/F_p until one has a rational point Q_0 of order 2^t, and pick R' ∈ E'(F_p) with 2R' ∉ ⟨Q⟩.
- For j = 0…t−1, take the kernel point K_j = 2^{t−1−j}·Q_j, with ξ_j = x(K_j) and τ_j = 3ξ_j² + a'_j.

- The Vélu step is `x ↦ x + τ_j/(x − ξ_j)`, onto the codomain `a'_{j+1} = a'_j − 5τ_j`, `b'_{j+1} = b'_j − 7ξ_jτ_j`.
- With `c = x(Φ(R'))`, the equation `L(x) = c` has exactly the `2^t` rational roots `x(R' + ⟨Q⟩)`, and `V` is that set.
- Mixed radix (`degrees=2,2,3`) is a parameter.

Why it is the generic family. It places no condition on `p`. So on P-192 (`v₂(p − 1) = 1`) it is the route that remains. A Dickson torus tower (`2^t | p + 1`, `t ≤ 64`) also exists there. It was first measured as a tower in PR #1893, at two primes with `v₂(p − 1) = 1` (`T24`, `T31`): it gives the same `D`, width and steps as the split Dickson tower in all 16 cells, and it is degenerate at infinity like the split towers (A5c). So on a P-192-shape prime the torus tower is the split computation on different numbers, and the isogeny tower stays the only non-degenerate family. The PKM note §4.5 (lines 600–603) makes this the only family that can support a statement about generic primes: "Conclusions about generic primes come only from `kind = isogeny`." It is also the one tower that is not degenerate at infinity (§3.4).

Measured (only at `p₀ = 786433`, `t ≤ 10`). The pilot, round 2 and the A1 table give these values of `D`:

`m`	`N`	isogeny `D`	width	Kummer `D` at the same `N`
2	2…18 (pilot, `f4_fp`)	4,5,5,5,5,5,5,5,5	9 … 17,754	4,4,4,4,4,5,5,5,5
2	2…20 (round 2, raw presentation)	3 5 5 5 5 5 5 5 5 6	35,158 at `N = 20`	3 4 4 4 4 5 5 5 5 6
3	6, 9, 12	6, 6, 6	399; 3,543; 13,249	5, 6, 6
4	8, 12	6, 7	1,897; 21,841	6, 7

- "It rises at the same level as Kummer's, `t = 10`" (line 1423). The A1 reading for the isogeny `m = 2` cell is inconclusive (line 1511).
- Generator-count ladder at `N = 12` (instrument power): isogeny `D` = 5, 5, 5, 5, 4 for `g = 1, 2, 4, 8, 12`. The instrument registers a fall, but the isogeny system resists adding generators longer than Kummer does.

The auxiliary-curve search. These counts are the `aux_curves_tried` field of `research/pkm_tower_pilot_20260924/runs/*.jsonl` (naive `O(p)` counting at `p₀`):

t	1	2	3	4	5	6	7	8	9	10
curves tried	1	3–4	1–5	8–54	88	3–142	196	404	584	425

The `E'` found had order 786,688 at `t = 8` and 787,456 at `t = 10`. The pilot reported this search but did not charge it. The design rule (§2.1, §5.4) lists "charging the auxiliary-curve search as free" as inadmissible, so any priced run must add it.

Derived in the first version of this note: at P-192 size with `m = 4`, `t ≈ 48`, so the note's own estimate (line 246) is about `2^48` SEA point counts per factor base. PKM's cheaper alternative builds `E'` by CM through `x² − Dy² = 4N` (requirements README lines 110–119). It was not implemented. Correction (2026-10-16): `t ≈ 48` is `N / m` with `N ≈ 192`, the size at which every point decomposes; it is not the cost optimum. The note's own §3.7 puts the optimal base at `|V| ≈ (m!·r·T)^{1/(m+1)} ≈ 2^39` at `m = 4` (derived), i.e. `t ≈ 39`.

Charged end to end in ecbench (2026-10-15, PR #1889, `research/pkm_isogeny_tower_ecbench_20261015/README.md`). The base is now the `ic.pipeline` plug-in and `ecbench fb` family `prime-isogeny:t=T[,seed=S]`, with the auxiliary-curve search charged to the factor-base phase in group-addition equivalents. 1,080 runs on eight curves of 20–28 bits, three of them with `v₂(p − 1) = 1` (the P-192 shape); every run verified and 720/720 replays identical.

- Measured. Index calculus on the isogeny base costs **1.002–1.331× the same index calculus on a plain base of equal size** (paired `S` ratios; p24 is unresolved at 1.002 [0.979, 1.028]). The excess is the search's share of `S`, **0.8% to 25%**, set by how many curves the seed happened to try (3 to 744), not by curve size; it is 13–25% on the three `v₂(p − 1) = 1` curves. Every IC arm stays far above rho (isogeny ÷ rho 21.0–136.5).
- Measured. Curves tried ≈ `C·2^t` with pooled `C = 1.05` over 224 builds (`t = 8…14`), with a spread of two to three orders of magnitude between seeds at fixed `t`.
- Derived, not measured. At P-192 size (`t = 48`, `m = 4`) the search costs about **`2^74` GAE** with a `2^25`-entry baby-step table (`2^82` at `t = 64`), against rho's `√(πr/4) ≈ 2^95.8`. So the search would not dominate; the relation and linear-algebra phases, which the solving-degree question decides, would.
- Verdict (class: accounting). The isogeny tower buys a table oracle nothing over a plain base of the same size, as the Kummer tower did not; the price of its construction is now inside `S`. No exponent claim.

The auxiliary curve at the P-192 prime: 2-Sylow halving (2026-10-15, PR #1925, `research/pkm_isogeny_tower_p192_20261015/README.md`).

- Built. `src/cryptanalysis/two_sylow_search.rs` and the binary `pkm-aux-search` search for `E'/F_p` at `p = 2^192 − 2^64 − 1` without counting points. Each draw is a curve with full rational 2-torsion, whose 2-Sylow is `Z/2 × Z/2^b`; x-only point halving computes `b`.
- Measured, cost per draw (P-192, 65,536 draws × 7 interleaved repetitions): on the `Z/2 × Z/8` family (`z2xz8`), the median is 30.83 µs for the baseline and **10.06 µs** for the `fast` arm, **0.326×** the baseline. All four pre-registered speed claims pass. Shared host, so the timings are noisy.
- Measured, depth distribution. `2^28` `z2xz8` draws (45.6 min on one core) and `2^25` `z2xz2` draws match the pre-registered Haar model (`P(b ≥ t) = 2^(3–t)` for `z2xz8`) at every tested `t`; draws to success were 8,161 ± 45 at `t = 16` (model 8,192).
- Found and checked. Ordinary curves with points of order `2^16`, `2^24`, `2^28` and `2^38` (the deepest of the `2^28` draws). `backlog_check_aux2t` accepts each and rejects `t + 1`. On the `2^38` curve the **Vélu tower ∨ is built at P-192** for `T = 8, 12, 16` (`|V| = 2^T` distinct abscissae; `L(x) = c` holds on all of `V`, in the field backend and over `BigUint`).
- Derived: *`t = 48`*. About `2^45` `z2xz8` draws × 10.06 µs, i.e. **98,352 core-hours ≈ 11.2 core-years** (range 95,617–99,624). It extrapolates the Haar model 17 bits past the deepest tested `t`.
- The shortcut (measured). Over this prime `y² = x³ + x` is supersingular, with cyclic group of order `p + 1` and `v₂(p + 1) = 64`. `pkm-aux-search --family supersingular-j1728 --t 48` returns a point of order exactly `2^48` from the first draw, in 7 ms (also `2^64`); the checker accepts both and the tower builds on that curve (`T = 8, 16`). Whether PKM's relation phase may use a supersingular `E'` was left open, and PR #1947 tested it (next block).

A supersingular auxiliary curve (2026-10-15, PR #1947, `research/pkm_supersingular_aux_20261015/README.md`). The plug-in takes `prime-isogeny:…,aux=supersingular` (families `ss-j1728`, its twist, `ss-j0`, `ss-j0-twist`). Three pre-registered questions:

- Q1, structure of *`V`* (exact): not Möbius-equivalent to the torus; formally "partial". Over 103 primes `p ≡ 3 mod 4` (103–2999) and `t = 3, 4, 5`, no `y² = x³ + x` set and no `j = 0` set is `PGL₂(F_p)`-equivalent to a Dickson torus set. Two of 12,969 `y² = x³ − x` sets match one torus set, at one cell (`p = 383, t = 3`), which the frozen rule counts as partial. Ordinary curves: 0 of 24,580. So the supersingular route does not inherit the torus verdict.
- Q2, F4 solving degree (measured): supersingular *`D`* = ordinary *`D`*, and both ≠ torus. In every one of the study's 14 `(m, N)` cells, at each of up to three primes (24–31 bits), for all four supersingular curves: `D = 3, 5, 5, 5, 5, 5, 5, 5, 5` at `m = 2, N = 2…18` and 6 at `N = 20`, with widths equal to the ordinary tower's within

one column. The torus gives 4 at `N = 4…10` and 5 against 6 at `m = 3, N = 6`. H2a holds, H2b fails, so H_same fails, as the protocol predicted.

- Q3, charged relation phase (measured, ecbench at 20–24 bits, 504/504 verified). The auxiliary work is **0.9–1.9% of s** with `aux=supersingular` (the build alone), against **10–27%** for the searched ordinary curve, with unchanged yield. Under the §5a band (0, 0.1, 1 GAE):
  - supersingular ÷ ordinary on s22 **0.82 / 0.84 / 0.92** and on s24 **0.74 / 0.76 / 0.88**: cheaper at every weight, so this ranking holds;
  - on s20 **1.11 / 1.10 / 1.07**: dearer at every weight (its base is 15% larger), so H3c fails there;
  - supersingular ÷ plain base of the same size: unresolved under §5a on s20 and s24 (the order flips inside the band, point ratios 0.98–1.02) and 1.01–1.07× dearer on s22, so "costs the same as a plain base" is not held;
  - every IC arm is 29–94× rho at `w = 0` and 36–85× at `w = 1` (lower-bound comparisons on both sides; rho is cheaper at every weight).
- Conclusion, as the study states it. "The 11-core-year ordinary search buys nothing these experiments can see": a supersingular `E'` (one draw) is a sufficient input for the relation phase. It does not touch lead 1, whether `D` stays bounded as `N` grows, which remains the gate.

Status. Open on the solver axis; the solver measured at 20–31-bit primes only, `N ≤ 20`. Priced end to end in ecbench at 20–28 bits (PRs #1889, #1947; accounting: no gain over a plain base for a table oracle). At P-192 the auxiliary curve and `V` are built (PR #1925), and the supersingular curve makes the auxiliary search free (PR #1947); no F4 run or relation phase has run at P-192, because the plug-in and the isogeny system builder are `u64`. The autoresearcher program cites the construction (`AR:KN-LIT-025`) but never built it: `AR:H-RELN-96e3ba` calls the PKM arm "unmaterialised". The P-192 `t = 48` auxiliary search card (§8) is abandoned as superseded (2026-10-16 claim audit): `t = 48` is not the §3.7 optimum (`|V| ≈ 2^39` at `m = 4`), and the supersingular `E'` and the Dickson torus make the search unnecessary.

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1. How each entry reads

Each entry gives as many of these fields as the record supports:

- **Definition**: the set `V` of abscissae, or the set of points.
- **Existence**: conditions on `p`, `n` or the curve.
- **Build**: how the base is constructed, what it costs, and whether that cost is charged.

- **Fold**: the symmetry quotiented out (negation, Frobenius, automorphisms, 2-torsion).
- **Oracle**: the decomposition oracle or solver paired with the base.
- **Measured**: the results, with the sizes they were measured at.
- **Verdict**: the recorded class or decision (refuted, rejected_scoped, weakened, inconclusive, engineering, accounting, open), with the mechanism when the source names one.
- **Sources**: where the record lives.

Families are grouped by field regime: §2 prime fields, §3 odd-characteristic extension fields, §4 binary fields, §5 Koblitz curves, §6 Jacobians and covers. §7 collects the results that bound every family at once, and §8 lists the open lines.

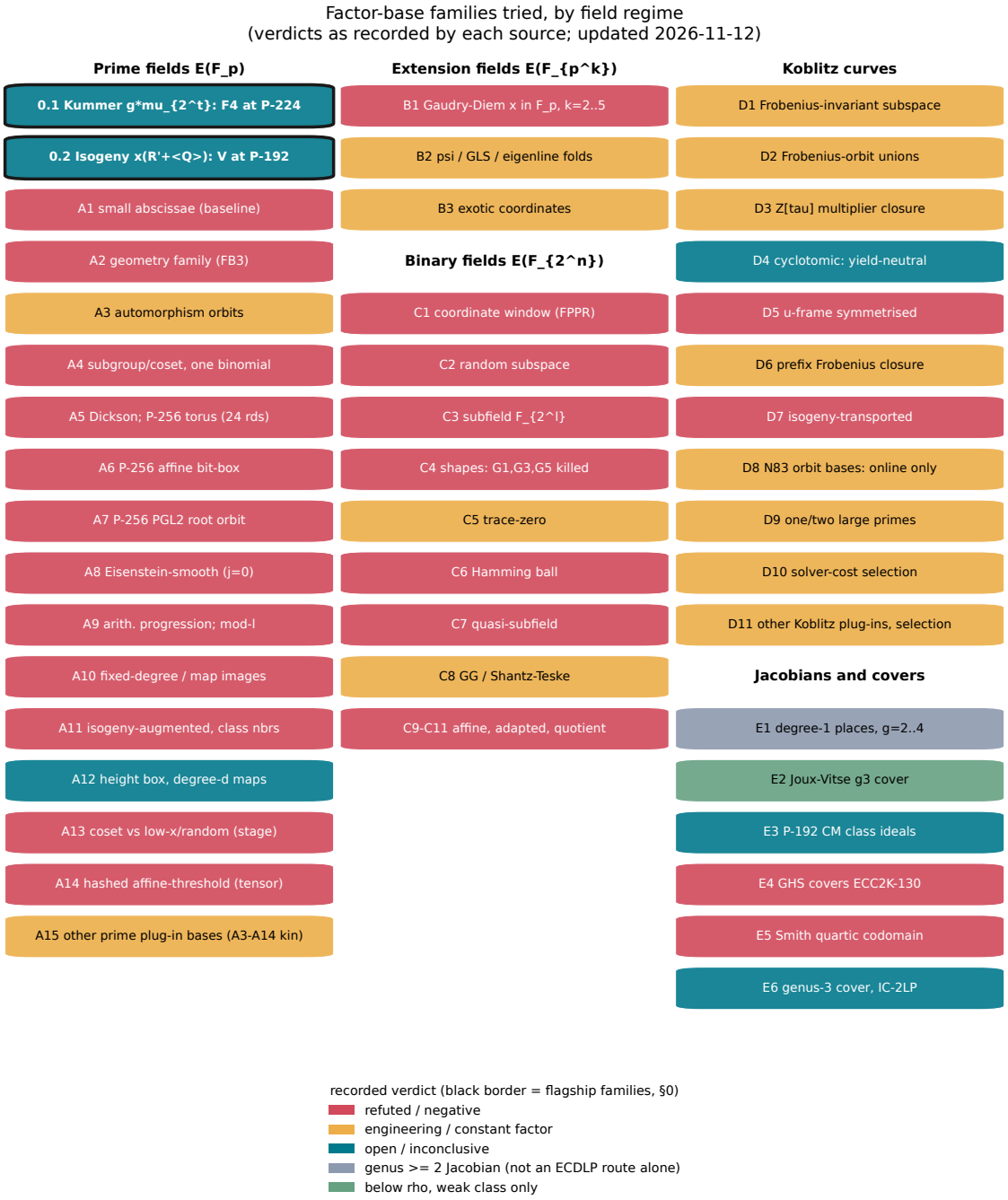


Figure 1: Taxonomy of the factor-base families, coloured by recorded verdict

2. Prime fields $E(\mathbb{F}_p)$

A1. Small abscissae / x-interval (Semaev's prime-field base): the baseline

- **Definition.** `F = {P : x(P) < B}`, both signs, one column per abscissa. `|F|` is about `#E^{1/3}` for `m = 3`.
  - The membership polynomial `∏(x − x_j)` is dense, of degree `|F|`.
  - Variants: a stride-`s` scan, a centred interval, the high-bit window `[p − W, p)` (`AR:H018`), and an offset interval.
- **Build.** One Legendre symbol and one square root per abscissa, charged.
  - Plug-in: `prime-abscissa:size=N[,stride=s]` (`src/cryptanalysis/ic_framework/plugins.rs` L71–133).
  - In the autoresearcher engine: `small_x` (`AR:src/crypto_autoresearcher/index_calculus/factor_base.py`).
- **Oracles.** Semaev `S₃` roots, `subtract`, `mitm` (`m = 2, 3`), `mitm-prime-batch`, `rootfind-*`, `S₄` with Cantor–Zassenhaus, Gröbner/msolve, crossbred.
- **Measured.**
  - Boundary ledger §2 (`research/notes/index-calculus/RESEARCH_IC_BOUNDARY_LEDGER.md`), eight sizes from `2^9.7` to `2^23.4`. At `r = 2^23.4`, `|F| = 512`: `S₃` roots 1,107× rho, subtraction 436×, `mitm m = 2` 59.1×, `mitm m = 3` 14.5×. The fitted exponents are 0.905, 0.850, 0.662 and 0.601, against rho's 0.274. Yield over ceiling is 0.75–1.25: the base "behaves like a random set of its size".
  - Deployed shape (`a = −3`, prime order; `RESEARCH_ECC2K130_IC_FEASIBILITY.md` §5): IC/rho is 19× at 16 bits, 114–199× at 20 bits and 233–403× at 24 bits.
  - `AR:EXP-ALPF-022` (`m = 3`): IC/rho is 478.63, 531.07, 1211.60 and 2109.18 at 12, 14, 16 and 18 bits, with fit `log₂(IC/rho) ≈ 0.368·log₂ n + 4.42`. Verdict: NEGATIVE (scoped).
  - `AR:EXP-ALPF-032`: a verified end-to-end solve. The decomposition exponent is 0.889, and the linear algebra alone costs `n^{2/m}`, more than `√n` at `m = 3`, "even with a free decomposition oracle".
  - AR engine sweep, 12–32 bits:
    - exhaustive IC is about `N^{1.0}`, against rho's 0.50 [0.48, 0.52];
    - MITM `m = 3` gives 0.67 and `m = 7` gives 0.58;
    - at 32 bits, `m = 3` MITM is 34× rho;
    - msolve per target grows as `|F|^{3.80}` (`m = 2`) and `|F|^{4.41}` (`m = 3`).
  - ecbench L0 smoke, 20-bit prime, `prime-abscissa:size=32`: `S = 11.329` (12.3× the negation rho; "not a performance claim"; `research/ecbench_factor_bases_20261013/RESULTS.md`).
  - Small-root route (`RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` §2–3):
    - with `x ≤ p^δ`, a decomposition needs `δ ≥ 1/(2(m−1))`;
    - the Coppersmith reach is `1/(m(m+1)·2^{m−2})`: 1/6, 1/24 and 1/80 for `S₃`, `S₄` and `S₅`;

  - `m = 4` needs about `δ ≈ 1/5`, against a reach of 1/80.
- **Verdict.** Refuted as a route past rho.
  - The pair-table family law (§7) is `S = 0.75·r^{1/6}`.
  - `AR:KR-IC-5931ee`: "Naive Semaev index calculus over prime fields cannot beat generic methods".
  - This base is the control every other prime-field entry is compared with.

A2. Geometry family at matched size (`AR:H-FB3-001`, six geometries plus two priors)

- **Definitions** (`AR:experiments/EXP-FB3-001/amendment-001.yaml`).
  - `high_bit_interval`: the top window of `F_p`.
  - `small_height`: the smallest `max(|u|,|v|)` with `x = u/v mod p`.
  - `mixed_two_base`: an interval plus small multiples, with typed decompositions.
  - `coset_union`: "Choose the largest divisor `d` of `p − 1` with `d ≤ B`, let `H` be the subgroup of order `d` in `F_p^*`, and take the union of cosets `g^i·H`", keeping the abscissae of points.
  - `greedy_optimized`: submodular greedy selection from a pool of `4B`.
  - `asymmetric_sizing`: base sizes `B₁ ≥ B₂ ≥ B₃`.
  - The two priors: `H016`, a quadratic-residue base from an r-adding walk, and `H017`, small multiples `jG`.
- **Measured.** `N ≈ 2^14, 2^16, 2^18`; 12 prime-order curves; `m = 3`; exact counting over every target; 288 of 288 cells.
  - Mean-yield ratio is exactly **1.000000** for every untyped geometry (`high_bit_interval`, `small_height`, `coset_union`) at every size.
  - `mixed_two_base`: 0.3726, 0.3700, 0.3749, against a prediction of `6/16 = 0.375`.
  - `asymmetric_sizing`: 0.034–0.042.
  - `greedy_optimized` (held out): 0.955–0.981.
  - `coset_union` coverage reached 1.0012, with a CI that includes 1. Its `d` was as small as 4, "a union of 59 tiny cosets".
- **Mechanism** (`AR:knowledge/findings/KN-FIND-007.md`, established). `Σ_r c_D(r) = C(B+m−1, m)` holds for **every** base of size `B`. "Factor-base geometry cannot change mean yield, only redistribute it." Coverage headroom is at most `min(1, μ)/(1 − e^{−μ}) ≤ 1.582`. A Bose–Chowla Sidon `B₃` base reaches 1.1071.
- **Verdict.** rejected_scoped (`AR:DEC-20260724-007`, `AR:EV-FBG-001`, replicated).

A3. Automorphism-orbit bases (`glv-orbit`, j = 0 / j = 1728)

- **Definition.** The closure of the `size` smallest abscissae under `Aut(E)`.
  - `j = 0`: `x ↦ ζx`, 6 points per column.
  - `j = 1728`: `x ↦ −x`, 4 points per column.
  - Each orbit folds onto one column with eigenvalue weights.

  - Plug-in: `glv-orbit:size,group=auto|negation|j0|j1728` (`plugins.rs` L889–948).
  - Ancestors: `ec_index_calculus_j0.rs` here, and `CA:prime_orbit_index_calculus.rs`.
- **Measured** (`RESEARCH_GLV_INVARIANT_FACTOR_BASES.md` E1 and E12).
  - The column ratio is exactly 3 (`j = 0`) and 2 (`j = 1728`).
  - The full-rank ratio is 2.83–5.48 and 2.22–2.96.
  - E12, a pair table over orbit representatives: the table shrinks ÷2.95 and `S` ÷1.18.
  - ecbench prime20 smoke: `S = 22.39`, against 11.33 for `prime-abscissa`.
  - `CA:`: 7–18× below per-target rho, but 0.72–0.89 of the folded batch-rho expectation.
  - `AR:H-STR-002`: an early 0.011 cost ratio disappeared once the appended rows were removed (`AR:EV-STR-003`).
- **Verdict.** Engineering: "The fold by any finite group of endomorphisms is the order of the subgroup of `(Z/rZ)^*` its eigenvalues generate". In other words, a constant `|Aut|`.

A4. Multiplicative subgroup and coset bases, high-degree form

(`V = g·μ_d`, one binomial) This is the same set as the Kummer tower (0.1), written as one binomial instead of a tower. It is the cleanest controlled test of whether the set helps without the presentation.

- **A4a. Sage subgroup base** (`research/ecdlp_autolab/structured_fb_subgroup_result.md`, 2026-05-27).
  - Cells: `x^m − 1` at 13, 16 and 19 bits, with `|FB ∩ H|` = 10, 16 and 31.
  - Gröbner time per query: 12.5, 17.3 and **74.7 ms**, against 4.7, 18.1 and 30.8 ms for a generic base. That is 2.4× slower at 19 bits.
  - Mechanism: the Hilbert regularity depends on `deg F = |FB|`, not on sparsity. This motivated Proposition 1 (§7).
- **A4b. AR engine `subgroup`.** `x ∈ g·μ_d` with `d | p − 1` chosen near `2|F|`; membership is `x^d = g^d`. A prime filter makes the comparison fair.
  - Exponent: 0.97 [0.95, 0.99] at `m = 2` and 1.00 [0.98, 1.01] at `m = 3`; "the high-degree subgroup base behaves like a random set of the same size".
  - msolve: the sparse binomial is about 4× faster than dense at `m = 2` (`|F| = 258`), but 20× slower at `m = 3` ("134 s against 7 s at `|F| = 25`").
- **A4c. Relation-level excess against matched random.** `AR:EV-PFDR-ad6274` and `AR:EV-PFDR-1faf10`, 30–32 bits.
  - `κ_rel` upper bounds are 1.0074–1.0112; the TT4 subgroup reads 1.0633.
  - One unreplicated excursion: z = 4.752.
  - Verdict: weakens.

- **A4d. Character-level.** AR:H-ECDLP-4e1880 (weakened, strong): coset indicators sit at the random Fourier level, e.g. index-3 coset 0.5013 ± 0.0037.
- **A4e. On paper** (RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md §4.6). Cyclic-norm folding in O(log B) products still yields a degree-2B output, so the per-target cost is Õ(B).
- **Verdict.** Refuted: no exponent change. The tower presentation of the same V (0.1) is the only open form. A13 compares the coset set with low-x, low-digit-weight and random bases of matched size under one instrument.

A5. Dickson bases

- **A5a. Toy.** x = u + c/u with u ∈ g·μ_d; membership is D_d(x, c) = λ (AR:FactorBase.dickson).
  - Indistinguishable from random: "dickson TT4 kappa 1.0172, upper 1.0886" (AR:EV-PFDR-ad6274).
  - As a tower (0.1) it is the same F4 computation as Kummer.
- **A5b. The P-256 Dickson-torus campaign** (24 rounds, 2026-10-04 to 2026-10-06; research/icv1-fp256-…_round{1..24}_*; builds on PR #1327). This is the only tower-type base built on a real NIST prime.
  - **Definition.** P-256 has $v_2(p+1) = 96$.
    - The generator: the norm-one element (1,5)^{p−1} in F_p[i]/(i² + 1), raised to (p+1)/2^96, gives an element of exact order 2^96.
    - The fibre is a depth-18 trace coset with terminal trace 0.
    - The abscissae are the 2^18 traces w + w^{−1}, the roots of the Dickson chain z_{j+1} = z_j² − 2.
    - The base is **131,239 negation-folded columns** (262,478 signed points), FB1h2ea06bef7f7a. The best inventory found was 131,458 columns (FB1h2f8621cda105).
  - **The round chain.** Most rounds ran on small-prime analogues (p = 127 or 1151); round 8 transferred to the real P-256 modulus.
    - Rounds 2–4: D = 4.
    - Round 5: branching 3–4 levels gives the first fall to **D = 3**, in 1,920 components.
    - Round 6: every extra level of residual depth multiplies the total work ×4.
    - Round 14: at real P-256 size the images reach the generic maximum (2, 8, 128, 32768), so there is no compression.
    - Round 18: the charged outer scan costs **2^272.404**, 144.4 bits above rho.
    - Round 19: a multilevel selector brings it to 2^165.156, 32.69 bits above rho. "The depth-18 chain removes zero global combinations."
    - Round 20: a batched collision gives 2^151.885, 19.419 bits above rho.

  - Round 21: **zero** FB1 relations in 8,413,312 small multiples and 5,986,344 Dickson-block sums.
  - Round 24: the selected Dickson base is **394.5× rho**. A K = 1 known-log row reaches 0.964×, but that is a lower bound, not an attack.
  - ▸ **Verdict.** A degree fall appears only after splitting into components, and the branch frontier grows exponentially. End to end it is refuted at 16–144 bits above rho.
- **A5c. The Dickson torus tower** (2026-10-15, PR #1893, `research/pkm_dickson_torus_20261015/README.md`). The A5b rounds built the torus set this is the first run of it as a PKM tower, `y_{j+1} = y_j² − 2`, `y_{t−1}² = 2 + c`, with `V` and `c` from `F_{p²}` and `2^t | p + 1`. Such a tower exists on P-192 (`t ≤ 64`) and P-256 (`t ≤ 96`).
  - ▸ Measured at `T24 = 7·2^21 − 1` and `T31 = 61·2^25 − 1` (both `v₂(p − 1) = 1`): **H_same**. The torus `D` equals the split Dickson `D` in all 16 cells (`m = 2`, `N = 2…20`; `m = 3`, `N = 6…15`; `m = 4`, `N = 8, 12`), with the same width and step count: it is the same F4 computation on different numbers. The positive control (isogeny at `p₀`) separated, and an exhaustive recount agreed on 127 of 127 rows.
  - ▸ Derived. The torus equations are the split ones with a different `c`, so their top forms are `y_{i,j}²` and the tower is degenerate at infinity (BFS degree of regularity `N + 1`).
  - ▸ **Verdict.** It inherits every open question of the split tower (0.1). It is not evidence that `D` is bounded: the `m = 2` line rises to 6 at `N = 20` exactly as Kummer's does.

A6. P-256 affine bit-box `x = offset + Σ_{i<18} 2^i·b_i`, with `b_i² = b_i`

- **Source.** `src/cryptanalysis/p256_bitbox_factor_base.rs`; round 1.
- **Measured.**
  - ▸ `S₃` solving degree is 5 against Dickson's 4 at depths 3–4, and 6 against 4 at depth 5.
  - ▸ The base `FB1ha8fa8dae90f9` has 131,440 columns.
- **Verdict.** Refuted ("reject degree and cost", round 22).

A7. P-256 PGL₂ root orbit: Möbius images `M(μ_d)` of subgroups of `F_p^×`

- **Definition.** `M ∈ PGL₂(F_p)` is drawn from a hash, and `d ∈ {260,270, 272,425}` divides `p − 1`. The base is closed under `u ↦ ωu`. There are 16 arms with 2,131,217 columns in total.

- **Measured** (round 23, `root_orbit_fb_round23_20261006/`). **Zero** cross-column relations in 34,099,472 images. The best arm, `FB1h8fc8b5fd8529`, is 491.494× rho.
- **Verdict.** Refuted: "generator degree two is not solving degree".

A8. Eisenstein-smooth base (j = 0)

- **Definition.** `x_α = (u + v·ω_p) mod p` for Eisenstein integers `α = u + vω` with `N(α) ≤ B²`; needs `p ≡ 1 (mod 3)` (`ec_index_calculus_j0.rs` L721–790).
- **Measured.** Exponent `α = 0.825` (`R² = 0.992`), against 0.752 for the `j = 0` orbit base and 0.691 for a generic base (`research/notes/ecdlp-general/RESEARCH_BENCH_LOG.md`).
- **Verdict.** "Empirically falsified … the speculative direction is dead."

A9. Arithmetic progression and mod-ℓ partitions

- **Arithmetic progression** (`research/ecdlp_autolab/structured_fb_additive_result.md`). Gröbner time is flat at 4.6–14.9 ms. The progression found was `x₀ = 0, d = 1`, which is just the generic base. Negative, with that methodology caveat. `AR:EXP-FB-001` (interval, progression, random and subgroup bases): "yields within 1.1x of random … solve time `~d^6` identical across structures".
- **mod-ℓ partition** (`research/ecdlp_autolab/phase9_result.md`, `E(F_1009)`). `x(P₁ + P₂) mod ℓ` is not a function of `(x₁ mod ℓ, x₂ mod ℓ)` for `ℓ ∈ {3, 5, 7, 11}`, over 19,801 pairs each. Structurally negative: the chord-tangent inversion does not commute with reduction mod ℓ.

A10. Fixed-degree and map-image bases (`AR`: ALPF series, 2026-05-30 to 2026-06-01)

- **Over `F_{p²}`: trace-zero, subfield and norm-1 bases** (`AR:EXP-ALPF-009`).
  - The membership degree is fixed and `D_reg` is about 7, where the interval base gives 10, 18, 34 and 66 at `L = 4, 8, 16, 32`.
  - But the fraction of relations landing in `E(F_p)` is 0.000 (trace-zero) and 0.007 (norm-1): relations land in the wrong subgroup.
  - Verdict: SCOPED_NEGATIVE, "descent blocked".
- **`E(F_p)`-native candidates** (`AR:EXP-ALPF-017`, on `y² = x³ + 10 / F_97`):
  - the CM eigenset is the whole group, because `[56]` is a scalar;
  - the **ℓ-isogeny image** is "BIJECTIVE … image = all of `E₂(F_p)`: no filtering";
  - `t² + c` needs a quadratic-residue test of degree 48;
  - `ψ_n` is empty on a prime-order curve.
  - Verdict: NEGATIVE.
- **`ψ_n` torsion base on `n² | #E`** (`AR:EXP-ALPF-024`). "n-torsion DLogs are all multiples of `n·L` … Relations carry zero info about `k mod L`." Pohlig–Hellman territory. FAILED.

- **Rational-map pullback $\varphi(T)$** (AR:EXP-ALPF-007, 011). D_reg falls from 18 to 7, but never both degrees at once. INCONCLUSIVE, because the meter was not validated.
- **e-ring and power-sum presentations** (AR:EXP-ALPF-011/013/021/031).
  - They fired in 16 of 16 cells.
  - The red team showed the effect is a "shared-factor syzygy living entirely in the FB constraints".
  - Verdict: FAILED (a bankable negative).
- **Torus T_2, Weil restriction, theta/Kummer chart** (AR:EXP-ALPF-018/023/025/026). None of them descends, or the solving degree is invariant under restriction. FAILED or INCONCLUSIVE.

A11. Isogeny-related prime-field bases (other than the tower)

- **Augmented $F \cup \varphi^{-1}(F)$** for Elkies $\ell \in \{3\ldots23\}$ at 20–28 bits (AR:H-ICI-063e91, weakened). The augmentation ratio is about $2 - B_frac$, the same as a random-bijection null ("$\Delta|aug - null| < 0.004$").
- **Isogeny-class neighbours** (AR:EXP-ALPF-029…031, AR:FINDING-PF-IC-001). $top_form(S_3)$ is independent of (a, b), and the solving degree does not differ.
- **Conductor-gap / fibre-aware bases** (research/isogeny_conductor_gap_ic_20261010/, research/drastic_prime_conductor_gap_*_20261012/). Floor/crater ratios are about 1 at gaps up to 2,033,609, with the same F4 degree (about 3) on both. Negative.
- **Yield label across isogeny classes** (ML:experiments/hardness/report.md, 106,594 nodes in 254 classes). The yield ratio γ/random has a within-class sd of 0.0021, and the best held-out R^2 is 0.021. The only within-class effect is mechanical: a factor-base 2-torsion point, worth about 2%.

A12. Proposals that are closed on paper or still at Stage 0

- **Height box** $F_H = \{P : ht(x(P)) \le H\}$ (AR:H-ECDLP-b97b72, specified). The relation search is pinned at the small-root threshold.
- **Degree-d map bases** $V = u_D^{-1}(W)$ (AR:H-ECDLP-30a2a1, AR:IDEA-20261009-a7733f). The design audit says "keep proposed and do not approve".
  - AR:KN-OPEN-020 (open) proves a bounded-degree no-go by derivation: by Bezout $B \le 3d$, at most B^m targets are reachable, and descent costs $\Omega(N/B^m)$.
- **CM-factor-base intakes** (AR:coordination/intake/cm-factor-base-*). Five diagnostic proposals, unapproved, with no exponent claim.
- **Global lifting** (xedni, small-height points of $E(Q)$), **embedding into $E(F_{p^n})$**, **quasi-subfield polynomials at $n = 1$**, and **Weil-descent analogues over F_p**: all closed on paper.

- RESEARCH_INDEX_CALCULUS_FACTOR_BASE.md §2 calls the last "structurally hopeless": the missing ingredient is Galois structure.

- **PKM divisor-set audit** (research/notes/cm-isogeny/RESEARCH_PKM_CRITERION.md). An audit score, not a base: P-224 and P-192 both score 0.647, and P-521 scores 0.001. The PKM note §1.3 says it is "neither … PKM's construction".

A13. Controlled comparison: multiplicative coset, low-x, low digit weight and random bases (PR #1898)

- **Definition.** Four point bases on the same prime-order curve, matched on the actual number of canonical ± point classes: a multiplicative coset (the PKM-style set, A4 and 0.1), low abscissae (A1), low digit weight, and random points. A separate quadratic-algebra norm solver after PKM §§3.3–3.4 was tried as a decomposition oracle.
- **Measured** (research/factor_base_evaluation_20261015/README.md, browsing copy in benchmarks/20261015-factor-base-evaluation/; 48 admitted curves over 24 fields at 16, 20 and 24 bits). At 24 bits, independent rows per total CPU second with the exact join: **low-x 1,352, coset 1,167, random 1,144, low digit weight 859**.
  - Coset ÷ random throughput **1.020**, with a descriptive 95% whole-field bootstrap interval of **[0.959, 1.083]**: "This does not establish a coset throughput advantage."
  - The norm solver passed its 24-bit positive-case gate, then produced 90 independent rows in **229.9 CPU s** on fresh inputs, against 115 rows in 0.268 CPU s for the exact join.
- **Caveats, as recorded.** A historical import: the run used Sage, Python and C, has no full-DLP solve, no matched rho, no counted end-to-end cost and no isolation receipt, and the repository requires a native rerun before it is extended. Stage throughput only, no ratio to rho, no §5a band.
- **Verdict.** Stage diagnostic. At matched size the coset set gives no measurable throughput advantage over random, consistent with yield conservation (A2, §7 item 1). The low-x ordering is descriptive and not band-checked.

A14. Hashed (affine-threshold) factor bases: Bochum v0.2 §3 and the tensor IC

- **Definition.** No set V of abscissae: m ordered slot bases $F_i = \{P \in \langle G\rangle : (\alpha_i \cdot \mathrm{encode}(P) + \beta_i) \bmod s < \tau\}$, s the first prime in $[2p, 4p]$, one independent (α_i, β_i) per slot, τ set so that $E|F_i| = \max(\lceil r^{1/m}\rceil, 10m)$ (Lemma 13). A relation is $R = P_1 + \ldots + P_m$ with $P_i \in F_i$. The ecdl-tensor-ic variant uses a keyed SHA-256 predicate by default, a random affine hash on the gadget path and

an optional coordinate window `x(A) < p >> k`. Membership is an affine map and a comparison, so it encodes as a low-degree predicate; there is no smoothness notion, so decomposition is an unstructured `m`-list problem.

- **Existence.** Any prime-field curve with a prime-order subgroup; a cofactor `c > 1` is handled by free source points `S` with `A = [c]S` (`ic.tensor_contraction_cofactor_v1`). The framework plug-in stops at `p < 2^61`; the vendored crate is arbitrary precision.
- **Build.** Inverse-hash enumeration: for each `t < τ`, decode `α^{-1}(t − β) mod s` to a point and keep it if it is in the subgroup. `O(τ)` decodes, no `O(p)` scan, no square-root table. Charged as curve additions plus `hashed_tensor_steps` and `sqrt_solves`; the tensor-curve certification is not charged, and `run_pipeline` says so in `unpriced_work`.
  - Plug-in: `hashed-affine:slots=m[,seed=…,tau=…,s=…,steps=…]` (`src/cryptanalysis/ic_framework/hashed.rs`), paired with `hashed-tensor` on the same parameters, which it refuses to run without.
  - Crates: `crates/hashed-ecdl-tensor` (Bochum §3 reduction, Appendix D.4 recursion, certificates), `crates/ecdl-tensor-ic` (whole-point network, ecbench `ic.tensor_contraction_v1` / `_cofactor_v1`, policy `DiagnosticOnly`), `crates/hashed-ecdl-cpu` (MITM oracle over the same slot bases, persistent tuple-sum tables, batch inversion).
- **Fold.** Negation only: one column per `{P, −P}`; the slots stay ordered.
- **Oracle.** Exact Boolean tensor contraction (`hashed-tensor`, the crate's `SparseContractor`); meet-in-the-middle over the slots (`hashed-ecdl-cpu`, `Solver::mitm`); plain enumeration in the bounded reference. Every witness is re-added in the group before a row is accepted.
- **Measured.** All `[pcc:unverified]`; nothing has reached the gate.
  - Counts-only protocol v2, 2026-10-31 (`research/hashed_ecdl_counts_20261031/`): `ic.tensor_contraction_v1`, `m = 2`, against `rho.negation` at 0, 0.1 and 1 GAE per unpriced operation: **$4.3{\cdot}10^3$–$9.5{\cdot}10^3\times$** at 10 bits, $2.7{\cdot}10^4$–$5.0{\cdot}10^4\times$ at 12, $1.7{\cdot}10^5$–$3.0{\cdot}10^5\times$ at 14, **$6.7{\cdot}10^5$–$1.2{\cdot}10^6\times$** at 16; the order holds across the band on every curve, and the ratio grows about 5.1× per two bits (log-log slope 1.17, 95% interval 1.00–1.34). Against the plain `prime-abscissa:size=32` + `mitm` baseline on the same curves: 259× to $2.2{\cdot}10^5\times$. The native hashed arms (`ic.hashed_native_v1`, branch `codex/hashed-ecdl-ecbench-e2e`) completed 3 of 16 cells, at $5{\cdot}10^2$–$2.8{\cdot}10^3\times$ rho; the rest exhausted their budgets and are unknown. The validator replayed all 416 executions and checked `[d]G = Q` on the 306 verified rows; one round, 8 targets per cell, A/A vacuous for counts.
  - CPU oracle, L0 shared host (`research/hashed_ecdl_cpu_20261030/`): `p = 65521`, `m = 3`, `τ = 80`: complete recovery median 45.99 ms → 3.90 ms with persistent

tables and batch-64 inversion, 120 executions per panel, byte-identical certificates across arms. No rho arm.
  - Desk audit, 2026-11-07 (research/hashed_ecdl_tensor_audit_20261107/): reduction correct; encoding $s = a \cdot m \cdot b$ edges with $a \geq 5.3 \cdot 10^3$ at $b = 131$; the recursion is admitted only for $\lambda \geq 2^{17}$ ($s \geq 2^{131072}$ edges), so every run so far is direct summation ($root_u = 0$); the composed exponent is above rho by ≥ 120 bits at $b = 128$–256 even with $a = 1$; linear algebra alone exceeds rho for every $m \leq 6$. §7 item 1 applies as written: the hashed base behaves like a random set of its size.
- **Verdict.** Refuted at every measurable size, and closed as an experimental lead (closed_obstructed; card research/pdp_strategy_evaluation_20261031/cards/e4.md). The asymptotic claim is conditional on OpenAI Theorem D.1 and stays a watch item: revisit on an explicit-constants D.1 whose saving starts below about 10^6 edges. The hash is an encoding device, not a yield device.
- **Sources.** ECDLP_RESEARCH_TECHNIQUES.md ("Hashed factor bases"), FRAMEWORK.md ("Hashed-affine / hashed-tensor"), cryptanalysis/docs/hashed-ecdl.md, research/hashed_ecdl_leads_20261030/, research/ic_prime_eth_leads_20261031/leads/L09/.

A15. Other prime-field plug-in bases without their own entry

(inventory of 2026-11-12) Bases that ship on main and had no entry above. Each line gives the definition, the build, the record and the verdict as the technique matrix (ECDLP_RESEARCH_TECHNIQUES.md §1) records them; all numbers [pcc:unverified], none gate-validated unless said.

- **cm-multiplier:size=K,norm=N[,mult=M]** (ic_framework/plugins.rs). $Z[\varphi]$ closure on a CM curve: $[a]P + [b]\varphi(P)$ for every $\alpha = a + b\varphi$ of norm $\leq N$ on the seed's column with weight $a + b\lambda$; the prime twin of D3. A precompute trade, the relation floor unchanged; no dedicated record (the Koblitz analogue reads 4.5× rho). Engineering.
- **cm-scalar-orbit** and the ecbench ic.cm_folded_* methods (ic_framework/cm_orbit.rs, ecbench/cm_folded.rs): columns seeds times length points under the CM scalar action for $D = -19$ or -43, a signed MITM2 in a hash-pinned native backend with independent Rust replay. Floor/crater 0.98 [0.87, 1.11], "not an admitted IC/rho comparison" (research/conductor_gap_cm_orbits_20261015/). Accounting.
- **known-log:size=B** and ic.known_log_mitm2_v1, ic.known_log_ksum_v1 (ecbench/known_log.rs): the base $\{\pm[s_i]G\}$ with known logarithms, so no matrix; a Bernstein–Lange-shaped table. Cheaper than the abscissa pipeline by 1.5 → 15.6× from 16 to 24 bits, but negation rho is cheaper at every size (rho/known-log s 0.43 at 24 bits; research/ecbench_known_log_targets_20261021/). Not an attack.

- **`prime-interval:t=T[,lo=L]`** and the digit-set, Edwards-`y`, `T0`-invariant and square-bits frames of `pkm_semaev.rs`: the sets the bit-membership system of `pkm-semaev` describes; frames are Möbius coordinates of one system. Frame choice moves F4 cost 3.3–4.7× at `k = 20` (`research/curve25519_frame_search_20261103/`, `VALIDATION.json` present but unable to pass for want of a same-unit rho row); square bits closed; the greedy digit set is an open, not engine-confirmed lead. Engineering.
- **`archive:store=…,manifest=…,symmetry=…`** (`ic_framework/archive.rs`): imports a content-addressed factory archive with certified negation, cubic, quartic or Frobenius symmetry and the `useful_rank` audit; the transport for the factor-base factory. "No candidate has been selected by the comparison" (`ARCHIVE_FACTOR_BASE_COMPARISON.md`). Instrument.
- **`explicit-x:xs=…`** (`ic_framework/explicit_base.rs`, ecbench only): an exact abscissa list, both signs. Instrument.
- **Prime-orbit index calculus** (`prime_orbit_index_calculus.rs`, `ic prime-orbit`, `ic.prime_orbit_relations_v1`): `m` full `Aut`-orbits, two-unknown relations solved as a gain graph, scaled curves to 40 bits; 5.42× negation rho on a 14-bit toy curve, 4.20× cubic rho on `j = 0`, "rho-led". Engineering, the ancestor of A3.
- **Small-x exact-yield instrument** (`prime_small_x.rs`, `secg_small_x.rs`, `ic prime`): the exact two-sum yield of the smallest-abscissa base on every SECG curve. Loses to rho by at least `2^40.9` in trials on every curve; P-192 yield `2^{−175}`–`2^{−165}` (`research/secg_small_x_20261014/`). Refuted, the A1 baseline at deployed size.
- **Coordinate frames** (`coordinate_search.rs`, `coordinate_quotients.rs`): the module-level mechanism behind B3; toy fields, constant factor.

3. Odd-characteristic extension fields **`E(F_{p^k})`**

B1. Gaudry–Diem subfield base **`{P : x(P) ∈ F_p}`**

The one textbook positive case: Weil restriction supplies the conjugate equations that `F_p` lacks.

- **`k = 2`, toy** (`RESEARCH_DIEM_DESCENT.md`): `E/F_25`; 18 of 25 targets decompose.
- **`k = 3`** (`RESEARCH_RESIDUAL_WALKS.md` §11; `gaudry_cubic.rs`).
  - MITM with a Weil-restricted S_3 pair test: 1,057× rho at `2^24.2`, rising to 2,240× at `2^33.1`.
  - With Macaulay and engineering, the S_4 solve constant is $C_3 = 0.876 \cdot 10^6$. The best plain variant is 528× rho.
  - The double-large-prime exponent is 0.424 ± 0.016, below 1/2, but at 1,989× rho; parity is extrapolated to lie past `2^230`.

- **k = 4** (gaudry_quartic.rs). $C_4 = 1.2 \cdot 10^{12}$ per S_5 solve. The linear algebra against rho is $r_\infty = 0.518 \pm 0.031$. Handover is at about 2^{151}.
- **Joux–Vitse (k–1)-point decompositions** (RESEARCH_RHO_PARITY_PROGRAMME.md, RESEARCH_K5_TORSION_JOUX_VITSE.md).
  - $k = 4$: flat at 6,945×.
  - $k = 5$: the Edwards 2-torsion symmetry cuts the solve constant 416×; parity is at 2^{233}.
- **ML: M9b**: 453 curves, $n = 2, 3$; no first fall in 5,592 of 5,592 rows. The yield follows the product law.
- **Verdict.** Bounded away from rho at machine size. "On generic curves no route closes below 2^{200}."

B2. Symmetry-folded extension bases

(RESEARCH_GLV_INDEX_CALCULUS.md; RESEARCH_GLV_INVARIANT_FACTOR_BASES.md E2–E18)

- **ψ-folded subspace on $j = 0$ $E(F_{p^3})$.** Every cost falls exactly ÷3.0, but the result is still 220–540× rho.
- **GLS ψ-stable line gls-line.** $x \in u \cdot s \cdot F_p$ on the GLS twist over F_{p^2}, 4 points per column; the full-rank ratio is 2.20.
- **Frobenius eigenline (E5).** The column ratio is 3, but relations fall only 1.2–1.6×, because the π-fixed $E(F_p)$ confines the folded rows.
- **FGHR Y-line with the τ_T 2-torsion fold (E13).** 12 columns per orbit and a degree-16 resultant oracle; the best arm is **267×** rho.
- **E18, $k = 4$ ψ fold.** $r_\infty = 0.138 \pm 0.008$, against 1.232 plain; handover at $2^{146.0}$.
- **D_4-symmetrised S_5 on the τ_T line, $k = 4$.** On a curve over F_{p^4} with a rational 2-torsion point T, FGHR's coordinate Y has $Y(P + T) = -Y(P)$. With the base $\{P : Y(P) \in F_p\}$, S_5 is invariant under the Weyl group D_4 (order 192), and the Bézout count falls from 4,096 to 512.
  - **C8, unweighted** (2026-10-13, PR #1785, research/gaudry_quartic_d4_20261013/README.md). Five unknowns $(\sigma_1, \sigma_2, \sigma_3, \sigma_4, \pi)$ with $\pi^2 = \sigma_4$, Macaulay degree 14, a 13,412 × 11,628 matrix: $C_4 = 2.535\text{–}2.552 \cdot 10^{10}$ at $p = 269$, **48× below** the plain form's $1.213 \cdot 10^{12}$, not the ≈ 630× projected. Handover 2^{135} in C8's own model.
  - **Weighted, four unknowns** (2026-10-15, PR #1895, research/d4_symmetrised_s5_20261015/README.md). $(\sigma_1, \sigma_2, \sigma_3, \pi)$ with weights $(2, 2, 2, 1)$, weighted degree 8, a 3,979 × 4,200 matrix. Measured: about **$1.41 \cdot 10^9$** F_p multiplications per solve, flat in p (medians 1.411, 1.406 and $1.437 \cdot 10^9$ at $p = 269$, 1033, 4093); **18× below C8** (paired medians 17.8–18.5, at least 7.7× on each of 47 paired targets); **≈ 850× below the quoted plain C_4** (medians 860, 863, 844), so the registered 630× projection is met. The lift-back costs at most $2 \cdot 10^{-4}$ of a solve. 101 of 102 targets closed and

matched brute force or meet-in-the-middle; the one degenerate target (p = 17, quotient dimension 514) is unsolved by both arms, reported as unsolved rather than wrong, and the verdict carries that caveat. Derived: the handover in E18's τ_T frame moves to **2^118.9**, 39 bits before the plain S_4 form's 2^157.9 and 17 bits before C8's 2^135.6. No relation collection, linear algebra or rho was run. The gap between C8's 48× and the projection was the presentation, not the symmetry.

- **The k = 4 bases in ecbench.** ic.k4_gaudry takes base = xline ({P : x(P) ∈ F_p}), yline-d4 ({P : Y(P) ∈ F_p}, columns {±P, ±(P + T)}), yline-d4-lp (with large primes) and yline-d4-ed (src/cryptanalysis/ecbench/methods.rs).
  - D_4 plus tuned double large primes (2026-10-14, research/ic_improve_20261014/01-d4-2lp/README.md, PR #1862): the whole solve **regresses, 1.025× [1.016, 1.037]** over 24 held-out curves (p = 101–1301), exponent +0.0055; the linear algebra alone falls to 0.827×, but it is a negligible share of the D_4 cost. ecbench (yline-d4-lp): 1.011× [1.001, 1.026]; untuned Gaudry-α 1.58×.
  - First-dependency stop (yline-d4-ed, 2026-10-15, research/ic_improve_20261015/04-d4-2lp-fix/README.md, PR #1862): **0.979× [0.973, 0.986]** of D_4 over 27 held-out curves (p = 101–1601), never worse on any curve, exponent unchanged; ecbench 0.990× [0.977, 0.998].
  - Both are paired ratios between two IC arms. The ecbench sessions label them "diagnostic only (unpriced work)", and neither was band-priced.
- **Q-curves (E14) and type-C CM endomorphisms (E4).** Base points survive only at chance rates, or the base has zero invariant points.
- **Verdict.** Engineering or accounting throughout: a fold divides by the group order and nothing more. The weighted D_4 solve is engineering too: a constant of one phase (the per-decomposition solve), with no exponent change.

B3. Exotic coordinates (Möbius frames, 2- and 3-torsion; RESEARCH_EXOTIC_COORDINATES.md)

- The S_4 degrees halve, from [4,4,4,4] to [2,2,2,2,1].
- On F_{31^3} the v-line base (86 points) gives the smallest system measured: 3 unknowns of degree 2.
- The Vélu and Klein transports are not F_p-valued.
- **Verdict.** Constant-factor.

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4. Binary fields E(F_{2^n}): subspace shapes

The master inventory is RESEARCH_FACTOR_BASE_SHAPE_SEARCH.md §1, table A1–A17, with the closure sweep in examples/factor_base_closure_sweep.rs.

C1. Coordinate window `V = ⟨1, z, …, z^{l−1}⟩` (FPPR /
Petit–Quisquater / Diem)

- **Plug-ins.** `binary-subspace:dimension=l`, alias `koblitz-standard-subspace`.
- **Oracles.** `descent-algebraic`, with 11 engines (F4/F5 variants, crossbred, XL, SAT-CDCL, FES, exhaustive), and the pair table.
- **Measured.**
  - Ledger char-2 rows: `mitm m = 3` is 89× rho at `n = 27` (exponent 0.624). The S_4 pairs-and-solve, a relabelling, is 45,471× rho.
  - ECC2K-130 product law: `2^132.58`, i.e. `2^71.77`× rho.
  - `AR:KN-FIND-aa2efc`: even a **free** oracle costs `2^89.25`, `2^68.58`, `2^56.40` and `2^48.44` at `m = 2…5`, against rho's `2^60.809`.
  - ecbench L0 smoke, Koblitz `n = 17`: `S = 43.10`, 47.2× rho-frob.
  - `CA:experiments/fb-search/`: a model-predicted selection over 1,850 bases has Spearman 0.924 against measured cost. The best base beats prefix `l = 5` by 2.8×, but "every measured cell remains 3.9e3–4.4e4 times the rho reference".
  - `CA:` PDP-degree heuristics: over 656 bases, `D = 2 + [e < −3.5] + [e < −8.5]`, where `e = n − dim V − dim V^{(2)}`. Geometric progressions are optimal (`dim V^{(2)} = 2l − 1`).
  - `CA:` linearized-half (Courtois 2016): `3l − 1 ≤ n` makes S_3 F_2-linear; on ECC2K-130 it costs `2^91.0` and up, +30 bits over rho.
  - **Binary MITM pair-table oracle** (2026-10-15/16; PR #1919 `research/binary_mitm_oracle_l8_20261015/REPORT.typ`, PR #1944 `research/binary_mitm_oracle_l9_l10_20261015/REPORT.typ`, PR #1964 `research/binary_mitm_oracle_l11_l12_20261016/REPORT.typ`). On a non-Koblitz curve (`a = z`, `b = 1`) with the base over `⟨1, z, …, z^{l−1}⟩`, all `P_i + P_j` are tabled once (charged to the factor-base stage), and a query looks up `R − P_k`. Against Gaudry pairs-and-solve over the same subspace, median per-target wall ratios: **94–277×** at `l = 8` (`n = 25`, three runs), **2,473.9×** at `l = 9`, **4,244.2×** at `l = 10`, 928.0× at `l = 11` and 2,187.2× at `l = 12` (`n = 37`; 259 MB table). The counted gap is the design's `2^l`: at most 16,092 point additions and lookups against 8,390,656 quartic solves at `l = 12`. Zero disagreements with pairs-and-solve on every target. Class: engineering, a stage `decomposition` record: the `2^{2l}` table replaces the `2^{2l}` per-target scan, so relation collection stays `0(2^n)`, and there is **no ratio to rho**. Wall times are from a shared, unpinned host.
- **Verdict.** Refuted at deployed size. "No engine here beats exhaustive search, and nothing beats rho" (`CA:suite/docs/ic/FRAMEWORK.md` §8).

C2–C8. Other binary shapes

entry	definition	measured	verdict
C2 random subspace	random F_2-subspace	D^* about 4.0 at $2l = n$, rising with l (DREG measurement note)	the hard baseline
C3 subfield F_{2^l}	$V = F_{2^l} \subset F_{2^n}$	D^* about 2.0; unavailable at prime n; on $b = 1$ the points lie in $E(F_{2^l})$ and the yield collapses	structurally empty at prime n
C4 EXP-H shapes	gp, gpscaled, gpsym, Frobenius progression, mixed, trace-zero variants	"No shape beats the coordinate window on class-0 targets"; at $m = 3$ (PR #1892) shape, modulus and Frobenius twist move the refutation degree by exactly zero at $l \leq 4$	negative; G1, G3, G5 killed; G2 survives at $n \leq 9$; G4 not built; G6 not run (below)
C5 trace-zero	$V \subset \ker \mathrm{Tr}$ (koblitz-trace-zero)	doubles the yield; class-1 targets have zero decompositions (AR:H-SEMBIN-83a856, 79,751 targets). As a pre-filter in the compact-orbit pipeline: **1.00003×**, because odd-order IC targets lie in $2E$ and it never	engineering

		fires (0 rejections in 29,624 checks; ic_improve_20261014/03, PR #1862)	
C6 Hamming ball	{wt_NB(x) ≤ w} in a normal basis	MultiSolve calls grow as |F_w|^{1.9–2.5} (hamming_ideal_pdp_20260930/RESULT.md: H1 refuted at n = 11…19; class accounting); CA: N53 weight-3: 3 DLPs, rho/IC 0.256–0.319	rejected (it saves storage only)
C7 quasi-subfield	X^{2^{n₀}} − λ(X)	"barren" at n = 131; AR:KN-FIND-617d78: β > 1/2	closed
C8 Galbraith–Gebregiyorgis / Shantz–Teske	special subspace; δ shift	GG "buys variables, not degree" ($D^* = 4$ at $n = 9$, ≥ 5 at $n = 15$); δ lowers D^* by about 1. The DoR campaign (factor_base_dor_campaign_20261013/REPORT.typ, PR #1826; Koblitz $n = 9…19$): F1, the subfield base (closed under products) refutes at $D^* = 2$ against 2.35–3.00 for standard and random bases; F10, shrinking l from 3 to 2 at $n = 15$ (δ +6 → +9) drops D^* from 5 to 3. At $n = 131$	constant factor

		there is no proper subfield, so only the δ lever exists there	
C9 affine plane at N19	12009 + span{257473, 353192}, 152 points	coverage 99.44% of the best; SAT 1.23× slower (AR admissible-affine study, 2026-09-21)	negative
C10 curve-adapted full-density subspaces	subspaces dense in x(E)	CA: density 0.079–0.994 of random at dimension 5; about dimension 10 at n = 131, against the 33 needed	negative
C11 orbit-valued quotient	Frobenius-orbit invariants as the base	full degree (n = 19: 18); missing entropy at n = 131	negative

Shape search G1–G6 at m = 3 (2026-10-15, PR #1892, research/shape_search_g1_g6_20261015/README.md). A stage diagnostic: refutation degrees of descended Semaev systems on toy binary fields (n ≤ 13), with no S and no ratio to rho. The goals are the ones frozen in RESEARCH_FACTOR_BASE_SHAPE_SEARCH.md §7.

- **G1** (shape gap at m = 3): **killed on the scaled cells** (5,2), (7,3), (11,4): gp, fp and coord sit exactly at random (gap 0.000). **Untestable on the frozen cells** (13,5), (17,6), (19,7): a single (13,5) probe did not finish in 60 minutes on the shared host.
- **G2** (τ-twisted chains): **survives at n ≤ 9**: rise 0.000 in every arm at (5,2), (7,3), (9,3). It survives a kill condition about degree, not about shape.
- **G3** (coordinate window across moduli): **killed**, panel spread 0.000.
- **G4** (trace-form, non-subspace Frobenius-stable sets): **not built**.
- **G5** (Möbius pole, m = 2): **killed**, the expected null (β-spread 0.06–0.19, under the 0.3 threshold).
- **G6**: **not run** no survivor of G1 or G3 to carry to n = 31.
- The positive control (the subfield F_8 ⊂ F_{2^9}) refutes at 5 where random reads 6, so the instrument sees a one-degree shape effect.

At these sizes the refutation degree is set by `l` alone: 5 at `l = 2`, 6 at `l = 3, 4`. The open question is `l ≥ 5`, where the ladder reads `≥ 7` and could separate shapes; this host could not reach it. The study also found that the shape-search note's frozen scale-8 `m = 2` run never landed: its evidence file is missing on `main`.

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5. Koblitz curves `K_a / F_{2^n}`

D1. Frobenius-invariant subspaces (`koblitz-orbit:divisor=i;j`)

- **Definition.** `V = ker` of a product of linearised factors of `x^n − 1`, i.e. a binary cyclic code. Signed Frobenius orbits fold onto one column with weights `±λ^k`.
- **Existence.** Dimensions are subset sums of 2-cyclotomic coset sizes. **The family is empty at `n = 131` and 163**: only dimensions 0, 1, `n − 1` and `n` exist.
- **Measured.**
  - Toys: relation savings 7.86–12.96×, linear-algebra savings 61.7–168×.
  - Ledger: K falls from 438 to 20 at `n = 23`, and `S` 7%. At `n = 41` it is 300× rho. This is the ledger's one "advance, count" label.
  - ecbench smoke `n = 17`: 18.7× rho-frob.
- **Verdict.** Engineering, and unavailable at deployed `n`. See `AR:KN-FIND-b9a41d` (the gain is at most `n`, and the only faithful stable dimension at 131 and 163 is `n − 1`).

D2. Frobenius-orbit unions (`compact-orbit-scan`)

- **Definition.** The first `K` cofactor-projected signed Frobenius orbits from a raw-x scan (`docs/ic/COMPACT_ORBIT_PRODUCER.md`, PR #714).
- **Measured.**
  - 5 columns give `S = 1.64` (1.55× rho).
  - Isolated `n = 61` online run: 12.9× and 13.9× faster than rho per target online, but the cold ratio is 83.4.
  - Every complete-CPU cold cell is above parity (1.314–22.675×).
  - ECC2K-130 total-work verdict: a materialised union bottoms out at `2^124.99`, i.e. `2^64.18`× rho.
  - ecbench `n = 17`: 9.985× rho-frob.
  - **Across the §5a band** (2026-10-15, PR #1837, `research/ic_frontier_campaign_20261015/README.md` 16 targets × 2 rounds per curve, every run verified, L0 counts). Mean `S` over signed-Frobenius rho at `w = 0` / 0.1 / 1 GAE per native operation: **3.8–15.0×** at `w = 0` and **11.8–30.3×** at `w = 1` over Koblitz `n = 13…47` ($\log_2 r$ 11–39); for example 3.8 / 4.9 / 11.8 at `n = 17` and 15.0 / 17.8 / 30.3 at `n = 41`. Fitted exponent at `n = 37–47`: 0.589 [0.518, 0.743] against rho's 0.436 [0.358, 0.508] at `w = 0`, separated there and overlapping

at `w = 0.1` and 1. Rho is cheaper on every curve at every weight. `ic-shared` leaves three categories uncounted, so the IC side is still a lower bound.
  - **Free relations** harvested from the root table (constant, paired against the same pipeline without them). Complete solve **0.957× [0.944, 0.970]** counted (`n = 41, 53, 61`; `ic_improve_20261014/03`) and, tuned against tuned on held-out curves, **0.960× [0.946, 0.976]** (`ic_improve_20261015/02`); both PR #1862. **Default on** in `examples/koblitz_orbit_dlp_fast_online.rs`; `KIC_FREE_RELATIONS=0` restores the old pipeline byte for byte. The trace-zero pre-filter tried with it never fires (C5). These are IC-against-IC ratios, not ratios to rho, and were not band-priced.
- **Verdict.** It wins online after precomputation only. Priced across the band, it stays 3.8–30.3× rho.

D3. `Z[τ]`-multiplier closure (`compact-orbit-scan:norm=N`)

- **Definition.** Orbits of `α(P)` for `α = a + bτ` of norm at most `N`.
- **Measured** (`research/ic_leads_20261010/RESULTS_ROUND2.md` L391–400).
  - 2 columns at norm 64 give **`s ≈ 0.94`, 4.5× the matched signed-Frobenius rho**, the best toy row on the ladder.
  - At `n = 41`, `m = 3`, one-target total work falls 9.2×.
- **Verdict.** Engineering, a "precompute trade, not a yield lever" (`docs/ic/RESEARCH_TRANSFER_AWARE_IC_20261010.md` §4).

D4. Cyclotomic γ-weight base (`cyclotomic:w`)

- **Definition.** `C_w = {Σ_{k∈S} γ^k : |S| = w}`, with `γ` a primitive `(2n+1)`-th root of unity.
- **Existence.** `2n + 1` prime and `ord(2) = n` (so `n` is one of 11, 23, 83, 131).
- **Measured.** ecbench `n = 23`, `a = 1`: `S = 3.427`, 9.9× rho. The solver cells timed out.
- **The falsifier's yield clause, settled** (2026-10-15, PR #1887, `research/cyclotomic_falsifier_20261015/README.md`). An exact count over all `r` targets at `n = 23` (`K_1` and `K_0`) of `|B+B|`, `|B+B+B|` and additive energy, against 1999 random Frobenius-stable bases with the same columns.
  - Measured. **Yield-neutral**: all 15 tests (10 coverage, 5 energy) lie in the null (smallest two-sided p 0.363 against a critical 0.00333; largest deviation 0.65%). The positive control, a λ-progression base, is detected on every `n = 23` cell (p = 0.001). The `n = 11` cells cannot separate structure from chance and do not count.
  - Measured, cost (ecbench, operations only, L0). Against compact-orbit-scan at equal columns, cyclotomic is **costlier** by **1.022** [1.021, 1.023] (`K_1`, `w = 2`) and **1.122** [1.120, 1.125] (`K_0`, `w = 2`), both from the factor-base build phase, and unresolved in the other two pairs. Every IC arm is 2.7–30× rho-frob.

  - Not tested. The solver clause (a membership system of lower solving degree) is untested; the only record, at `n = 11`, went the wrong way (F4 degree 4 against 3 for ONB). `n = 83` was not run: the plug-in is single-word and no exact cheap cell exists.
- **Verdict.** In yield, `C_w` behaves like a random Frobenius-stable base of the same size, and it costs slightly more to build. The lead survives only in the narrow sense that its yield is not lower; any value it has must come from the untested solver clause.

D5–D7. Other Koblitz bases

- **D5. u-frame symmetrised base** (`koblitz-symmetrised`, `F_u = {P : 1/(x+1) ∈ V}`).
  - `S_sym/S_x` is 0.84 at `n = 17` and 0.76 at `n = 23`, but 4.0 at `n = 31`.
  - Every arm is 990–63,500× rho.
  - Refuted at `n = 31`.
- **D6. Prefix-subspace Frobenius closure** (`koblitz-closure`). Hits per system are 1.79× and 2.14×, against predictions of 2 and 3. It costs 50–100× more than the pair table.
- **D7. Isogeny-transported and descendant-native bases** (A16, degree 7).
  - The native base costs 4.012× more.
  - Floor-curve twins are level-blind.
  - `CA:` volcano descendants: E0 needs 0.996× the attempts (CI 0.976–1.017).
  - Negative.

D8. The N83 Koblitz programme: signed-Frobenius orbit bases at `n = 83`

- **Instances.** Two different workloads share the degree and must not be merged (`research/n83_evaluation_20261012/README.md` "Exact fixture and identity correction"):
  - the **evaluation and tournament curve** `EC1N83Ckb1a1haacb69ee9f37`, `y² + xy = x³ + x² + 1` over `F_2[z]/(z^83 + z^7 + z^4 + z² + 1)`, with a prime subgroup of order 8,569,786,107,849,059 (about 52.9 bits) and cofactor 1,128,547,018;
  - the sweep's **primary** model `icv1-f2m83-tm6151469093347-debefd74`, `y² + xy = x³ + 1` (`a = 0`), with an 81-bit subgroup (`research/koblitz_n83_factor_base_sweep_20261008/README.md`).
- **Definition.** `K` signed Frobenius orbits (166 points each) of subgroup points, folded to `K` relation columns: `K = 600` (99,600 points) in the evaluation, `K = 3,240–3,596` (537,840–596,936 points) in the tournaments. The oracle is a precomputed table of the `S₃` roots over pairs of base points, queried so that each target gives a four-point relation. The variants are storage layouts of

that table: `root4`, `root5`, `root11`, `root21`, `root22` (bytes per root slot or state), a Bloom filter, a sparse linear-map canonical selector (`sparse-canonical`) and the `wide3` layout. The sweep's mixed-orbit supervisor and hosted paired gate landed in PRs #1739 and #1931.

- **Measured** (all wall clock, not the `s` unit; one cloud or local host per study; none band-priced):
  - Cold, K600 against strong rho (PRs #1719, #1722, `n83_evaluation_20261012` five paired repeats of one public target). Fresh-process **IC/rho 218.03 [172.36, 285.57]** prepared-online IC/rho 1.382 [1.036, 1.882]. IC took 1,054.6 s per fresh process (1,047.9 s of it reusable preparation; 5.8 GB peak) against 5.07 s for rho.
  - Online, Tier 1, K3240 *root21* (`n83_tier1_final_results_20261014/RESULT.md`, PR #1824; five unseen public points, one GitHub Actions allocation): median **rho/IC 54.10**, bootstrap interval **[17.18, 58.90]**, every relation and scalar replayed. The IC clock starts after base, index and factor logs are ready: setup, 580–640 s per process and rebuilt for each point, is excluded, against rho's 1.2–5.0 s per solve.
  - Selection panels (PRs #1868, #1872, #1904, #1906, #1910, #1912, with results in #1917; 1,000–20,000 fresh points each): the sparse canonical selector cut K3300/`root11` online time from 39.722 to 31.963 ms; K3300 beat K3250 (+2.534 ms [+1.796, +3.263]); K3300/`root11`/`wide3` led the `root4`, `root21` and `root22` challengers; K3425/`root4`/Bloom and K3450/`root11`/`wide3` entered the final as `BOUNDED_UNKNOWN`.
  - Final tournament (PR #1917, `n83_final_tournament_results_20261015/RESULT.md` 4,000 fresh points, mirrored order): K3425/`root4`/standard 28.69 ms, K3450/`root11`/`wide3` 28.81 ms, K3300/`root11`/`wide3` 30.67 ms, K3596/`root4`/off 35.38 ms. Decision **`BOUNDED_UNKNOWN`** between the first two. A same-point Tier 1 pair for each finalist (`n83_unresolved_tier1_results_20261015/RESULT.md`) gave rho/IC 413.3× and 63.9× online, one point each; "a single point cannot resolve the factor-base population ranking".
  - High-K confirmation (PR #1945, `n83_root4_highk_confirm_results_20261016/RESULT.md` 8,000 points): K3500/`root4` 27.56 ms against K3596 29.74 ms, `PANEL_SELECTED` for K3500, with a K3596 timing drift between its two processes (+6.43 ms) that the study flags for an A/A replication.
- **Verdict.** Engineering, online only. After a 10-minute, 10–15 GiB setup the prepared base answers a 53-bit-subgroup target in tens of milliseconds, 17–59× faster than strong rho in wall time, but the cold cost is 218× rho at K600, and the sweep's own requirement stands: "the requested minimum **complete cold index-calculus runtime remains unresolved**". The ratios

are wall-clock on shared or cloud hosts, not priced across the §5a band, and are not evidence about ECC2K-130 (AGENTS.md §8a: the subgroup is 53 bits). It is the D2 trade at a larger degree.

D9. Large-prime domain extension (one and two large primes)

- **Definition.** Beside the base `F`, a domain `D` of the next `lp_domain_factor·|F|` signed points from the compact-orbit scan. A partial relation leaves one or two summands in `D`; partials are merged exactly (`±λ^t` orbit coefficients mod `r`) and every merged row is replayed in the counted group before it reaches the matrix. Plug-in `mitm-frobenius:m=3,large_primes=2,lp_domain_factor=10` (`ic_framework::large_prime`); in ecbench, the method `ic.large_prime` on the `binary-nested-large-prime` family (`src/cryptanalysis/ecbench/methods.rs`), under `ECBENCH_LARGE_PRIME_PROTOCOL.md`.
- **Measured** (`research/ic_techniques_20261012/double_large_prime/README.md`, PR #1737; `K_0` at `n = 23, 31, 41`; 51 cells, 260 targets, every logarithm verified).
  - At `n = 23` and 31 (`r ≈ 2^21`) the plain table already decomposes almost every trial, and every large-prime cell is worse end to end (2.2–750×).
  - At `n = 41` (`r = 2^39`, 10 columns), trials per relation fall **8.7×, 36× and 138×** below the plain oracle with two large primes at `|D|/|F| = 3, 10, 30` (4.1×, 6.9×, 10.3× with one).
  - The best cell, two large primes at `|D| = 10|F|`, costs **17.7× the counted signed-Frobenius rho** for one cold target, against **420×** for the plain oracle on the same base. The equal-envelope control (a plain base as large as `F ∪ D`, 110 columns) reads 36.7×.
- **Verdict.** Engineering: a constant on a toy rung inside the pair-table family, and still above rho. The ratios are weight-0 counts, recorded before §5a: hash probes, canonicalisations and row operations are named but unpriced, so read them as lower-bound comparisons.

D10. Choosing among invariant subspaces by solver cost

- **Definition.** Not a new base: a selection rule among the admissible Frobenius-invariant subspaces of D1 (`koblitz-orbit:divisor=i;j`), scoring measured F4 cost instead of expected trials.
- **Measured** (`RESEARCH_FACTOR_BASE_SOLVE_COST.md`, evidence in `research/groebner_base_sweep_20260915/ icv1-f2m15-t275-b7f03703`, five bases, 60 verified logarithms). The trials-optimal selector ranks the bases almost backwards: **Spearman −0.90** against measured F4 word XORs, and the trials-optimal base (`2·3`) costs **12.12×** the cheapest. After a later change to the system builder, the matched re-run reads 11.48× and `ρ = −0.83` (§6 of the note).

- **Verdict.** A stage diagnostic (the Gröbner stage only, no `S`, no ratio to rho). The selector now scores solver cost when asked. At `n = 131` the question does not arise: the invariant lattice has dimensions `{0, 1, 130, 131}` only.

D11. Other Koblitz and binary plug-in bases and selection instruments (inventory of 2026-11-12)

- **`generator_orbit=1`** on `compact-orbit-scan`: column 0 is the generator's own orbit with known logarithm, one relation fewer. Nets 1.26–3.20× over the baseline on seven curves; at `n = 29` below the matched rho-frob across the band (`S` 0.546 / 0.611 / 1.196 against 1.569), which the record itself reads as a Frobenius-folded BSGS at one column on rho's fixed set-up; gap 1.15× at `n = 31`, 3.89× at 37 (`research/ecbench_generator_orbit_20261022/`, **validated**). Engineering.
- **`projected=1`** on `koblitz-orbit` and `koblitz-trace-zero`: folds columns by the signed orbits of the cofactor projections `[h]P`, the matrix's real column count; `no_fold=1` is the control. Accounting.
- **Greedy and sampled coverage selection, pruning, frozen `explicit_orbits` recipes, `subgroup_orbits`, 2-torsion saturation** (`koblitz_greedy_base.rs`, `koblitz_greedy_sampled.rs`, `koblitz_factor_base_search.rs`, `ic search`): expected trials at the best column count fall 1.08–1.18× at `r ≤ 2^17`, the selector's advantage over random falling to 1.000 at `r = 2^39` (`research/greedy_factor_base_yield_20261014/`). Engineering; past `r ≈ 2^28` use `subgroup`.
- **`binary-subspace … select=strict4e|gated-rank`** (`ic_framework/subspace_select.rs`): a sieved subspace with every abscissa in `4E`, or the best of several draws by a rank-aware key. Unresolved: 2.11 bits fewer counted operations at `w = 0`, the order flips inside the band, rho cheaper than both arms in every cell (`research/fb_strict4e_20261027/`, gate refuses its `VALIDATION.json`).
- **`binary-multiplier:size=K,mult=M`**: `[m]P` on the seed's column with weight `m`, structure-free; 6.1× cheaper than the best subspace base at degree 28 and still 6–12× rho (`research/ic_improve_20261023/`); "does not transfer". Engineering.
- **`explicit-x`** (binary): as A15. Instrument.
- **λ-coordinate base** `{P : λ(P) ∈ W}` (`koblitz_ic_leads.rs`, lead L1): a halved affine base, identity-checked on 182 subspaces, "L1 closed in full", nothing moves an IC cost (`research/koblitz_ic_leads_build_20261024/`). Closed.
- **Binary small-x / SECG base** (`binary_small_x.rs`, `ic prime --curve sect163r2`): yield $2B^2/(n-1)$, loses to rho by $\geq 2^{40.9}$ in trials (`research/secg_small_x_20261014/`). Refuted.
- **Shantz–Teske δ method** (`delta_method.rs`, Sage `--delta-sweep`): a dimension knob, $m \cdot l = n - \delta$; measured fractions track the prediction at `n = 31`, net cost per

relation 513 / 257 / 213 ms at δ = +3 / +1 / −1; "constant-factor tuning knob, no exponent change". Engineering (C8 has the clause).

- **Selection and forecasting instruments**: the planner (`ic plan`, `ic-planner/4`), the degree-of-regularity forecast (`dreg_forecast.rs`, `factor-base-sweep`), relation-yield pruning (`relation_yield_pruning.rs`), `ic fixed` and the 84 frozen `ic workflow --params` documents under `docs/ic/params/`. None carries an `S`; see the technique matrix §7.
- **N83 frozen bases**: D8 above.

6. Jacobians and covers

- **E1. Degree-1 places on genus 2–4 hyperelliptic Jacobians** (Adleman–DeMarrais–Huang / Gaudry; Thériault reduced base with large primes). `RESEARCH_HYPERELLIPTIC_IC_RHO.md`:
  - genus 2: IC/rho about 1;
  - genus 3: 0.66, 0.61 and **0.23** at `p = 61`, 101 and 401;
  - genus 4: 0.15 at `p = 61`;
  - "at reachable sizes the lever is the oracle … not the factor base".
  - These curves are not elliptic, so this is not an ECDLP route by itself.
- **E2. Joux–Vitse genus-3 cover base** on `Jac_H(F_{p²})` (`RESEARCH_COVER_DECOMPOSITION_LEDGER.md`).
  - `F = {(Q) − ∞ : x(Q) ∈ F_p}`, for the weak class of `E/F_{q³}` with a genus-3 GHS cover.
  - The sieve variant gives **0.039–0.069× rho at `p = 1009`** and **0.0122× at `p = 1511`**. From a non-weak curve, via an isogeny walk: 0.0049–0.019×.
  - **This is the only factor base in the inventory below rho, and only on the weak class.**
- **E3. P-192 weighted-CM class-relation census** (`research/p192-weak-curve-20261006/REPORT.md` L249–380; `AR:H-SCURVE-ec34c4`).
  - The base is prime ideals of `O_D`, `D = t² − 4p`, split `ℓ ∈ {13, 23, …, 113}`.
  - 9,948,061 vectors gave 103 relations, all scalar.
  - `INDETERMINATE`. It is the only factor-base object ever built on P-192, and it is not an x-coordinate base.
- **E4. GHS / Weil-descent covers of ECC2K-130.** The large horn is at best `2^70.19`× rho; the small horn is empty. `CA:` ECC2K-130 G2 gives `m = 1` or 131, so zero bits are gained.
- **E5. Smith plane-quartic codomain** (`ML:experiments/smith-codomain/report.md`). Diem line relations yield like a generic quartic: `f²/2`, and 10–30× the hyperelliptic yield.
- **E6. Genus-3 cover base after cubic GHS, with double large primes (IC-2LP).** Transfer the ECDLP on `E0(F_{q³})` (a structured class: `#E0(F_{q³}) ≡`

0 mod 4, split arm) to the Jacobian of a genus-3 hyperelliptic curve over F_q, and run index calculus there with the degree-1 places as base and two large primes.

- First run (2026-10-14, research/ic_next10_20261014/07-cubic-ghs-genus3/README.md, PR #1862; a Python tool with explicit counters, q = 2^8…2^15, 24/24 solves verified). Fitted exponent in q: **1.300 [1.292, 1.308]** over all sizes and **1.377 [1.363, 1.392]** over q = 2^11…2^15, against rho's 1.5. Against the strongest rho the instrument supports (measured cost per step × √(πℓ/2)), IC-2LP is **still 1.62× more expensive at q = 2^15**, and 2.29× against an ideal negation-map rho; a crossover is only extrapolated (q ≈ 2^17.2…2^19.2, about 2^25 once linear algebra mod ℓ is priced). Measured at weight 0 in F_q-multiplication equivalents, before §5a.
- Rust port and improvement rounds (IC against IC, paired, held-out curves): on a common range 2^10…2^18 IC-2LP has q^1.385 against the full base's q^1.910 (ic_improve_20261014/04); structured Gaussian elimination and Wiedemann give 0.653× at 2^20 (ic_improve_20261015/01); block Wiedemann 0.945× at 2^21 (ic_improve_20261015/06) (all PR #1862); cheaper per-trial primitives and a re-tuned κ 0.617× at 2^22 (ic_improve3_20261015/01, PR #1891); a Lucas splitting test plus lockstep walks 0.715× at 2^22, with the **local exponent at 2^21…2^22 down to 1.356** against the incumbent's 1.408 (ic_improve4_20261015/01, PR #1956).
- Verdict. Open. The exponent is below rho's 1.5 on this structured class but has not settled at the theoretical 4/3; linear algebra (about q^1.52–1.54 locally) is the remaining excess. No size measured is below rho, and no ratio to rho was recorded after the first run or priced across the band. Like E2 it applies to a weak class only, not to a generic curve.

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7. Results that bound every family

1. **Yield conservation** (AR:KN-FIND-007). Mean decomposition yield depends only on |F|, so no geometry changes it; coverage headroom is at most 1.582. This closes every "cleverer set of the same size" proposal that relies on a table oracle.
2. **Sparsity invariance, Proposition 1** (RESEARCH_INDEX_CALCULUS_FACTOR_BASE.md §1.2). For a fixed set V, every generating set of ⟨F_V⟩ has the same Hilbert function and the same D*. Re-encoding membership cannot get below the Yokoyama bound Reg ≈ md + d_S − m. **Its scope excludes towers.** PKM adds auxiliary variables (PKM note §1.3), so this is the exact gap the two flagship families (§0) live in.

3. **The pair-table family law** (boundary ledger §11.2–11.3). `S_family = 0.75·r^{1/6}` on a prime-order curve, which crosses rho only at `r ≈ 22`; about $1.58 \cdot 10^6\times$ rho at `2^128`. Base size and folds change only the constant. This is why A2, A4, the ecbench `prime-kummer` run and every table-oracle row read as "behaves like a random set".
4. **Symmetry inventory** (`RESEARCH_IC_LEADS_AND_CONNECTIONS_20261010.md` §6). Every fold (negation, Frobenius, automorphisms, 2-torsion, S_m, D_4) is a constant or polynomial in `n`. $S \cdot T^2 \gtrsim r/|\mathrm{Aut}|$ covers every table-lookup decomposition.
5. **Bezout bound** (`AR:KN-OPEN-020`). A membership predicate of degree `d` holds at most `3d` points, so a base of `B` points needs degree at least `B/3`. Only an auxiliary-variable presentation, a tower, escapes this.
6. **Subgroup confinement** (`AR:EXP-ALPF-009/024`, `CA:`/AR subfield rows). A base inside a proper subgroup, or in the wrong group after descent, carries no information about the target's logarithm.
7. **"Index calculus is priced by the ambient field; rho is priced by the subgroup"** (`RESEARCH_ECC2K130_DECOMPOSITION.md` §5.2), and "Prime `p` and prime `n` are one obstruction" (leads note §7(d)).
8. **The band-priced frontier** (2026-10-15, PR #1837, `research/ic_frontier_campaign_20261015/README.md`). One campaign ran every family on fixed curve ladders and priced every arm's unpriced work at 0, 0.1 and 1 GAE per operation (AGENTS.md §5a). Mean `S` over the cheapest generic arm on the same curve, at `w = 0` and at `w = 1`: Koblitz compact orbit 3.8–15.0× and 11.8–30.3× (`n = 13…47`); prime pair table 9.8 → 95.2× and 13.5 → 131.4× from 20 to 40 bits; PKM Kummer tower 30.3 → 141.3× and 31.0 → 238.7× from 20 to 28 bits. Fitted exponents: prime pair table 0.620 [0.557, 0.663] against rho's 0.435 [0.331, 0.501] at `w = 0`; compact orbit 0.589 [0.518, 0.743] against 0.436 [0.358, 0.508]; every other IC and rho pair overlaps. "No index-calculus arm beats generic rho on any curve at any band weight; no index-calculus arm has a fitted cost exponent below rho's." L0 counts only, and `ic-shared` leaves three categories uncounted, so the IC side is still a lower bound.

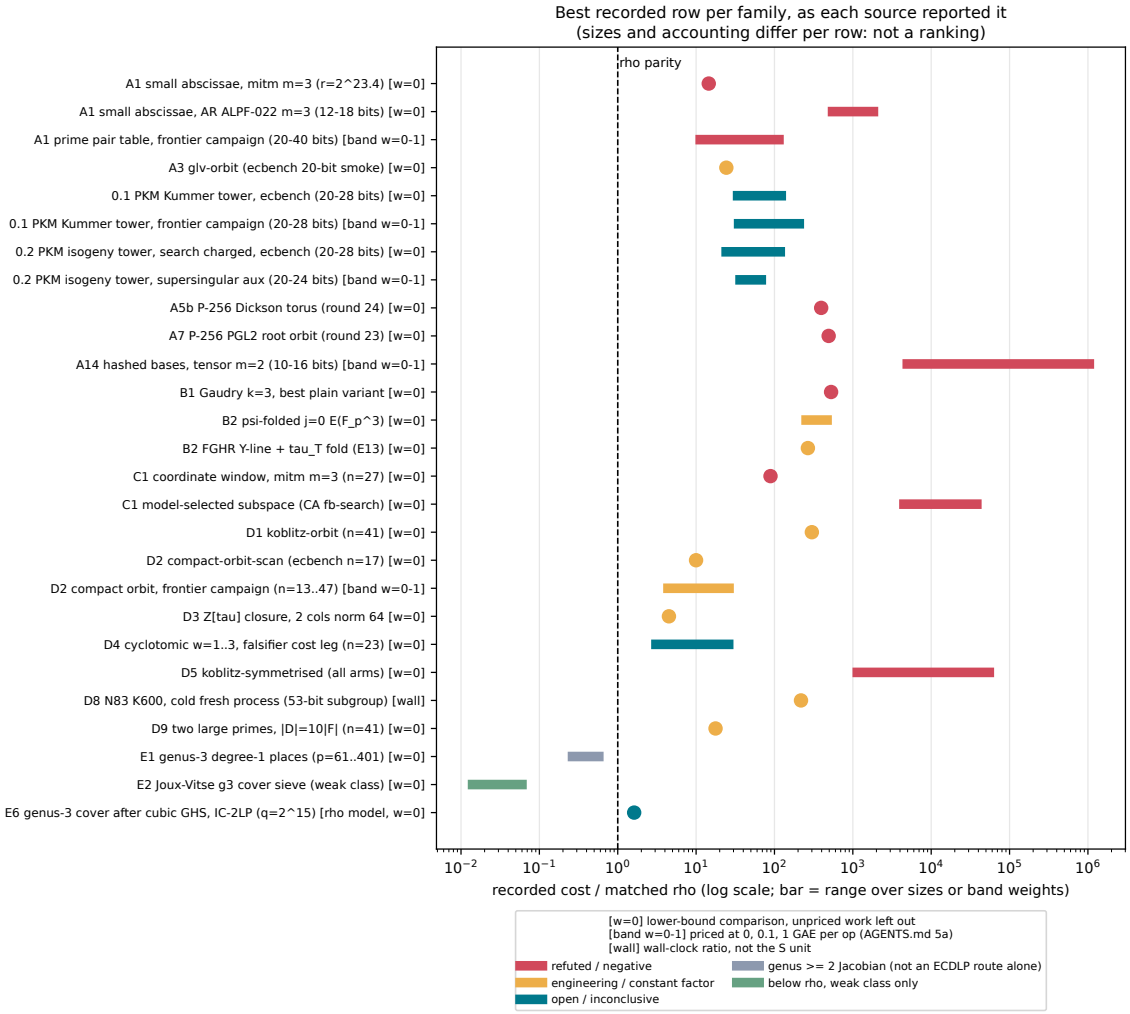


Figure 2: Recorded cost of the best row of each family relative to its matched rho (log scale; sizes differ by row)

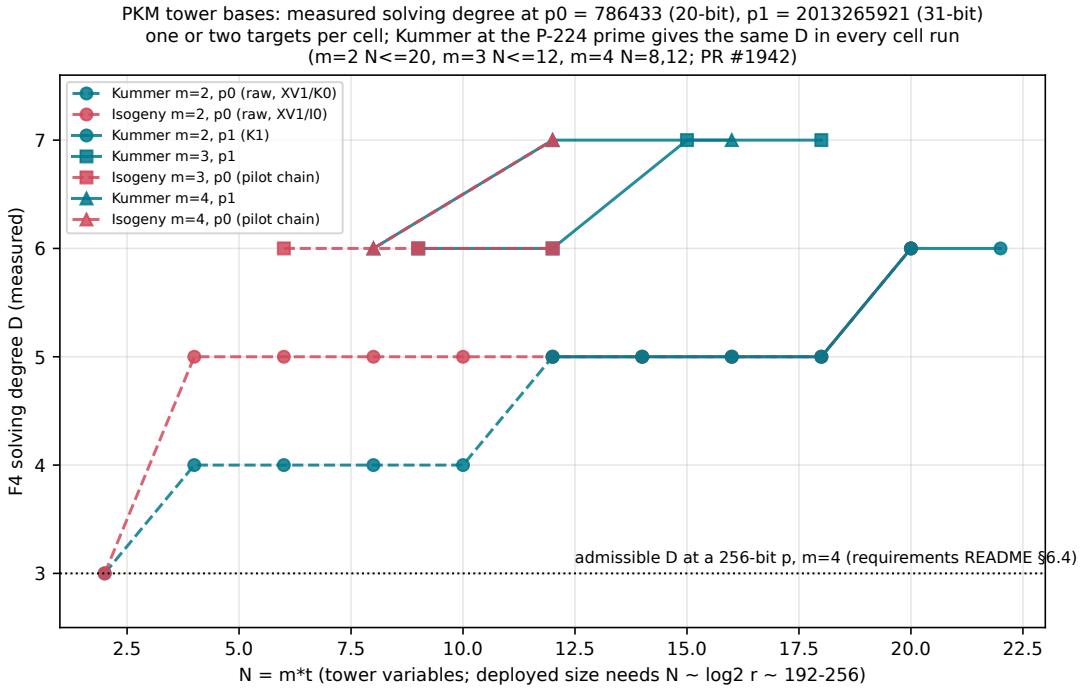


Figure 3: PKM tower solving degree D against N = m·t, Kummer and isogeny families

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8. Open lines, by how directly they test the two flagship families

The first version of this note listed five leads. All five were chased on 2026-10-15, and each item below gives its verdict and the merged study; items 2 and 4 were extended on 2026-10-16 (PRs #1925, #1942, #1947). The parts that are still open stay open. Item 6 lists the open lines of the entries added in this round.

1. **The $m \geq 4$ PKM solving degree past $N = 16$.** This decides whether either tower family is subexponential at all. **Answered at $m = 4$, NEGATIVE: $D \geq 8$ at $N = 20$** (`research/ic_next10_20261014/03-pkm-degree/`, 2026-10-14; msolve, raw presentation). Correction (2026-10-16): this item first read "still open" and did not cite that result. Only the $m = 3$, $N = 21$ cell remains open (inconclusive at $D \geq 7$), costed, not run (PR #1893, `research/pkm_dickson_torus_20261015/` §5). Derived: $m = 3$, $N = 21$ needs about 130–150 GB peak memory and $m = 4$, $N = 20$ about 240–380 GB on the current engine, so a 192–256 GB and a 512 GB host, or a disk-backed reducer block that is not built (the $m = 4$ projection assumed D stays at 7 and is moot). This container has about 14 GB.

2. **An isogeny-tower plug-in that is charged — done at toy size; verdict accounting** (PR #1889, `research/pkm_isogeny_tower_ecbench_20261015/`). `prime-isogeny` is in ecbench with the auxiliary search charged: 1.002–1.331× a plain base of equal size, search share 0.8–25%, curves tried ≈ 1.05·2^t. It buys a table oracle nothing over a plain base. Derived: the P-192 `t = 48` search costs about `2^74` GAE against rho's `2^95.8`; but `t = 48` is not the attack-optimal size (§3.7 puts `|V| ≈ 2^39` at `m = 4`), and the supersingular `E'` and the Dickson torus remove the search at P-192 (see the backlog card below).
   - The P-192 auxiliary curve — **search built and measured; `t = 48` derived at 11.2 core-years** (PR #1925, `research/pkm_isogeny_tower_p192_20261015/` §0.2). 2-Sylow halving at the real P-192 prime: `fast` arm 10.06 µs per `z2xz8` draw, 0.326× the baseline; the Haar model holds; ordinary curves with points of order `2^16`, `2^24`, `2^28` and `2^38` found; the Vélu tower `V` built at `T = 8, 12, 16`. `t = 48` needs about `2^45` draws, **98,352 core-hours ≈ 11.2 core-years** (derived). The supersingular $y^2 = x^3 + x$ gives a point of order `2^48` in one draw.
   - A supersingular E' — **sufficient for the relation phase at the sizes tested** (PR #1947, `research/pkm_supersingular_aux_20261015/` §0.2). `aux=supersingular`: its `V` is not Möbius-equivalent to the torus set (formally "partial": 2 of 12,969 $y^2 = x^3 - x$ sets match one torus set, at `p = 383, t = 3`); its F4 `D` equals the ordinary tower's in every cell, and both differ from the torus; its auxiliary work is 0.9–1.9% of `S` against 10–27%. Across the §5a band it is cheaper than the ordinary arm on s22 (0.82 / 0.84 / 0.92 at `w = 0, 0.1, 1`) and s24 (0.74 / 0.76 / 0.88), dearer on s20 (1.11 / 1.10 / 1.07), and against a plain base of its size unresolved on two curves and not held on the third. "The 11-core-year ordinary search buys nothing these experiments can see."
   - **Still open:** an F4 run and a relation phase at P-192 (the isogeny system builder and the plug-in are `u64`), and the decision the backlog card records: whether a supersingular `E'` is acceptable (the card is abandoned as superseded since the 2026-10-16 claim audit). Lead 1 remains the gate for all of it.
3. **The Dickson torus case** (`2^t | p + 1`) — **measured; H_same** (PR #1893, `research/pkm_dickson_torus_20261015/` A5c). The torus tower, which exists on P-192 (`t ≤ 64`) and P-256 (`t ≤ 96`), gives the same `D`, width and steps as the split Dickson tower in all 16 cells at two 24- and 31-bit primes, and is degenerate at infinity. It is closed as a separate question and inherits item 1.
4. **A tower at a prime whose $v_2(p-1)$ is P-224-like — measured; H_same, including the P-224 prime** (PR #1893 for $v_2 = 30$; PR #1942, `research/`

f4_generic_field_20261015/, §0.1, for the rest). The generic-field F4 engine runs Kummer towers at $v_2(p-1) = 40, 48, 56$ and at the P-224 prime ($v_2 = 96$), and the Dickson torus at P-192 and P-256. In all 82 cells (74 grid cells and 8 extension cells, among them Kummer $m = 2, N = 20$ and $m = 4, N = 12$ at P-224 and torus $m = 2, N = 20$ at P-192) D, width and steps equal the recorded 31–32-bit values. **Still open:** the second target in the one-target extension cells and $m = 3, N = 15$ at P-224 (the kummer-p224-depth card), and lead 1. The isogeny tower still has no F4 run at P-192.

5. **The three smaller leads.**
   - Binary shape-search G1–G6 — **G1 killed on the scaled cells, G3 and G5 killed, G2 survives at $n \le 9$, G4 not built, G6 not run** (PR #1892, research/shape_search_g1_g6_20261015/ §4). G1's frozen cells (13,5), (17,6), (19,7) were not testable on this host. The refutation degree is set by l alone at $l \le 4$. **Still open:** $l \ge 5$, the frozen G1 cells, and G4 (trace-form, non-subspace Frobenius-stable sets).
   - The cyclotomic base falsifier (D4) — **yield-neutral at $n = 23$** (PR #1887, research/cyclotomic_falsifier_20261015/). All 15 yield tests sit in the null and the positive control is detected; cost against compact-orbit-scan is 1.022× (K_1) and 1.122× (K_0). **Still open:** the solver-degree clause, untested, and $n = 83$, not run.
   - The D_4-symmetrised S_5 (Bézout count 4096 → 512). **Correction:** the first version said "never run". C8 ran it in PR #1785 (research/gaudry_quartic_d4_20261013/), unweighted, 48× below the plain form. The **weighted** four-unknown solve (PR #1895, research/d4_symmetrised_s5_20261015/ B2) costs about $1.41 \cdot 10^9$ F_p multiplications, 18× below C8 and ≈ 850× below the quoted plain C_4: the 630× projection is met. One degenerate target in 102 is unsolved by both arms. Derived: handover 2^118.9. Engineering; no $k = 4$ run below rho exists.
6. **Open lines of the entries added in this round.**
   - N83 complete cold runtime (D8). The online tournaments are BOUNDED_UNKNOWN between K3425/root4 and K3450/root11/wide3; the high-K confirmation asks for an A/A replication of K3596's timing drift; the minimum complete cold IC runtime is unresolved, and no N83 ratio has been priced across the §5a band.
   - Genus-3 cover linear algebra (E6). The local exponent is 1.356 at $q = 2^{22}$ against the theoretical 4/3; linear algebra, at about q^1.52–1.54, is the remaining excess, and 2^23 is out of the current memory cap. No ratio to rho after the first run.

- Binary MITM oracle (C1). An FFD-logged algebraic control on the same systems and an `l = 13` probe (a table of about 1 GiB) are the study's next rungs; no ratio to rho exists.
- Large primes at larger n (D9). Only the `n = 41` rung has headroom; its 5- and 20-column sweeps were not run, and the ratios are not band-priced.

The experiment backlog. `backlog/README.md` (PR #1903) holds runnable cards for the heavy runs that are still open. Their `status` on `main` at `05e414f3f` (from each `card.json`), with the 2026-10-16 claim audit's changes to two cards applied:

- `XP-20261015-pkm-m3-n21-m4-n20`: item 1's `m = 3, N = 21` cell. **Blocked** on hardware (192–256 GB) or a disk-backed reducer that is not built; rescoped 2026-10-16, since the `m = 4, N = 20` cell is answered (`D ≥ 8`);
- `XP-20261015-kummer-p224-depth`: item 4's remaining cells. **Draft**, with no blocker: the generic-field engine exists, and the card waits for a frozen `PROTOCOL.md`. Its status note prices `m = 3, N = 15` at P-224 at about $3.2 \cdot 10^{12}$ multiply-adds, about 13 h of one core;
- `XP-20261015-shape-g1-frozen-cells`: G1's frozen cells. **Blocked** on memory (47–50 GB at `(13,5)`, more beyond) and on the merge commit of PR #1892 being pinned;
- `XP-20261015-p192-isogeny-aux-search-t48`: item 2's P-192 auxiliary-curve search at `t = 48`, **abandoned (superseded)** by the 2026-10-16 claim audit: `t = 48` is not attack-optimal (§3.7: `|V| ≈ 2^39` at `m = 4`), its argv's `z2xz2` family contradicted the `z2xz8` estimate, and the supersingular $y^2 = x^3 + x$ and the Dickson torus give the tower at P-192 without a search. It had been **blocked** on a decision: whether PKM's relation phase may use the supersingular `E'` (then the search is unnecessary) or the checker must require an ordinary curve (then the card runs at about 98,000 core-hours). PR #1947's answer, at the sizes it tested, is that the supersingular curve is sufficient. The card also names lead 1 as its gate.

Two cards still name PR #1892 or PR #1893 as "open, unmerged" in their `blocked_on` text; both PRs have since merged. The cards' remaining blockers are hardware and, for the shape card, pinning the merge commit.

9. Method and limits of this note

- **Search.** Three read-only sweeps on 2026-10-15:
  - `crypto`: `research/notes/index-calculus/`, `docs/ic/`, `docs/ecbench/`, the `ic_framework` plug-ins, and `research/*` study directories;
  - `crypto-autoresearcher`: `knowledge/`, `ledger/`, `experiments/`, `coordination/`;
  - `cryptanalysis` and `ml-cryptanalysis`: the IC suite and the experiment reports.
- **Spot checks.** The key figures were checked against their files by hand:

  - the §10.10 2-adic table;
  - the pilot and A1 degree tables;
  - the PKM ecbench ratios;
  - the P-256 round-18 `2^272.404`;
  - the cover sieve's 0.0122;
  - the `Z[τ] S ≈ 0.94`;
  - the C4 table;
  - the `EXP-FB3-001` definitions and ratios;
  - `ALPF-022`;
  - the AR engine's subgroup exponents.
- **Update round** (2026-10-15, at `main b906df024`). The five lead verdicts in §0.1, §0.2, A5c, B2, C4, D4 and §8 were checked number by number against the study READMEs on `main` (PRs #1887, #1889, #1892, #1893, #1895), against the C8 README (PR #1785), and against the backlog cards' `status` and `blocked_on` fields (PR #1903). The figures were redrawn: the taxonomy labels for C4 and D4 and two `cost_vs_rho` rows (the charged isogeny tower; the cyclotomic falsifier's cost leg) changed.
- **Second update round** (2026-10-16, at `main 05e414f3f`). An audit of the second version found stale text in §0, §0.1, §0.2 and §8 and families that were missing. Every number added was checked against the cited study on `main`: the READMEs of PRs #1925, #1942, #1947, #1837 and #1898; the binary MITM reports (PRs #1919, #1944, #1964); the N83 RESULT files (PRs #1719, #1824, #1868, #1872, #1917, #1945); the `ic_next10` and `ic_improve` rounds (PRs #1862, #1891, #1956); the double-large-prime README (PR #1737); the solve-cost note; the DoR campaign report (PR #1826); the Hamming-ideal RESULT; and the backlog cards' `status`, `status_note` and `blocked_on`. Ratios are now labelled by pricing weight (§5a; see the conventions at the top). The figures changed: the taxonomy gained A13, D8, D9, D10 and E6, and `cost_vs_rho` gained band-priced rows (frontier campaign, supersingular auxiliary curve) and weight-0 or wall-clock rows (N83 cold, large primes, genus-3 cover), each labelled with its pricing.
- **Third update round** (2026-11-12). Added A14, the hashed (affine-threshold) factor bases of Bochum v0.2 §3 and the tensor IC crates, from the counts-only record of 2026-10-31, the CPU panels of 2026-10-30 and the desk audit of 2026-11-07; one taxonomy box and one cost row. The `research-kb` server and its CLI were unavailable in the authoring container, so the sources were read from the tree directly.
- **Fourth update round** (2026-11-12). Added A15 and D11, catch-all entries for every plug-in base on `main` that had no family here, from the module-by-

module inventory behind the technique matrix revision of the same date; two taxonomy boxes, no new cost row (their records are not mutually comparable).

- **What is not re-derived.** Entries condensed from long notes keep their source's status label and are not re-derived here. If a source and this page disagree, the source wins.
- **Nothing here is a new measurement**, and nothing here is a statement about a deployed curve. At P-192 and P-224 size only the F4 solving degree (P-224 Kummer, P-192 and P-256 torus) and the isogeny tower's construction (P-192) have run; no factor base has a priced relation phase there. The P-192 numbers `2^74`, `2^95.8` and 11.2 core-years, and the `k = 4` handover `2^118.9`, are derived by their studies, not measured.
- **Renderer.** Pandoc 3.1.3 converts this file to Typst, and Typst (Python wheel 0.15.0) compiles the PDF. The figures are drawn by `figures/plot.py` from `figures/data.json`, which holds only quoted values.
- **Conductor.** The control plane was not reachable in the authoring container, in the first version and again in both updates (`connection refused` on `localhost:8080`), so no task was reserved.